Scientific Reports

https://doi.org/10.1038/s41598-026-65256-3

Article in Press

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Received: 8 March 2026

Accepted: 30 July 2026

Published online: 07 August 2026

Cite this article as: Ruan J., Wang M., Peng L. *et al.* Mobility-aware predictive task offloading using deep reinforcement learning for future 6G-inspired vehicular networks. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-65256-3

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Mobility-Aware Predictive Task Offloading Using Deep Reinforcement Learning for Future 6G-inspired vehicular networks

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ABSTRACT

Reliable communication and ultra-low latency data processing are essential requirements for emerging Internet of Vehicles (IoV) applications such as cooperative autonomous driving, augmented reality navigation, and real-time safety monitoring. Although mobile edge computing has been widely adopted to reduce computational delays by offloading tasks from vehicles to nearby roadside infrastructure, the highly dynamic mobility of vehicles and rapidly changing wireless channels pose significant challenges for reliable task transmission and execution. Traditional offloading strategies typically rely on instantaneous network conditions and reactive decision mechanisms, which often lead to unstable communication links, frequent handovers, and increased energy consumption. This paper proposes a mobility-aware predictive task offloading framework designed for reliable edge communications in future 6G-inspired vehicular network environments. The proposed framework integrates trajectory prediction, wireless channel forecasting, and Deep Reinforcement Learning to proactively determine optimal task transmission and execution locations. Specifically, a Predictive Offloading Deep Reinforcement Learning (PODRL) architecture is developed, where predicted vehicle mobility and channel conditions are incorporated into an augmented Markov Decision Process (MDP) for intelligent decision making. In addition, predictive transmission path selection and proactive task migration mechanisms are introduced to ensure stable communication during high-speed vehicular mobility. In comparison to current reactive and reinforcement learning-based baselines, the proposed PODRL framework achieves up to 40% lower end-to-end latency ($p < 0.001$) and up to 45% lower vehicular energy consumption ($p < 0.001$) while maintaining high deadline satisfaction rates, according to extensive simulation-based evaluations conducted using realistic vehicular mobility traces and forward-looking 6G-inspired channel models. All evaluations are based on simulation-based 6G-inspired communication models and planned IMT-2030 capabilities, rather than on deployed commercial 6G infrastructure.

Keywords: vehicular communications, Internet of Vehicles, predictive task offloading, 6G-inspired vehicular networks, Deep Reinforcement Learning, vehicular edge computing, mobility-aware communication.

1. Introduction

The offloading of computationally demanding operations from resource-constrained vehicles to edge or cloud infrastructures is crucial to meet the rigorous latency and reliability requirements of modern vehicular networks. The Internet of Vehicles (IoV) enables real-time communication among vehicles, roadside equipment and cloud services which is a fundamental enabler of intelligent transportation systems [1]. These systems facilitate data-intensive and safety-critical applications such as cooperative perception and autonomous driving which need high processing speed and solid communication reliability [2]. For large-scale deployments, scalable vehicular communication infrastructures and efficient data exchange methods are required [3]. Recent frameworks highlight the integration of cognitive processing and distributed intelligence in IoV ecosystems [4,5].

High mobility and regularly changing topologies are the fundamental characteristics of vehicle environment, which make it technical wording to sustain reliable communication and decision making. Recently, predictive and mobility-aware routing schemes have been studied to optimize route planning and overall network efficiency for connected vehicles [6], [7]. In the meanwhile, collaborative perception techniques have been proposed to improve sensing coverage and reduce uncertainty through the sharing of information between vehicles and infrastructure [8]. Also, innovative transmission algorithms have been developed to improve the communication dependability under dynamic traffic conditions [9].

Dynamic task offloading and resource management have been widely investigated to adapt to fluctuating network loads, traffic patterns and computing demands [10], [11]. Some predictive offloading systems,

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such as OptiFog, rely on system state estimation to predict resource availability. Probabilistic trajectory models consider vehicle motion dynamics in order to enable proactive decisions [12], [13]. Finally, in the context of next-generation vehicle networks, intelligent resource allocation algorithms have been proposed to jointly optimize latency and energy usage under strong QoS restrictions [14].

Beyond mobility prediction, proactive network optimization is made possible by traffic-aware learning. For the purpose of estimating future states, spatiotemporal learning models have proven successful in capturing intricate traffic patterns [15]. In order to make adaptive and priority-aware decisions, multi-agent reinforcement learning has also been used for communication resource scheduling in vehicular networks [16]. While heterogeneous network optimization techniques further increase resource utilization efficiency [18], more sophisticated frameworks integrate distributed orchestration and service-level objectives into the offloading process [17].

In order to improve decision accuracy in dynamic situations, recent research emphasize the significance of non-periodic feature modeling and scalable traffic prediction [19]. In vehicular edge computing, Deep Reinforcement Learning (DRL) has been extensively used for task offloading, allowing agents to develop adaptive policies through interaction with the environment [20]. Advanced vehicular coordination and control algorithms reflect the complexity of real-world mobility patterns [22], while energy-aware scheduling mechanisms further improve system efficiency [21]. Furthermore, federated multi-agent DRL techniques allow for dispersed and privacy-preserving decision-making [24], and new routing solutions for integrated communication environments show the growing heterogeneity of next-generation networks [23]. Despite these developments, a significant drawback still exists: the majority of DRL-based offloading techniques currently in use mostly rely on real-time environmental observations, which might result in erratic decisions when there is high mobility and quick channel changes. While perception-driven learning approaches highlight the significance of contextual awareness in dynamic situations [26], mobility-aware offloading frameworks partially solve this problem by integrating trajectory information [25]. The advantages of anticipatory resource management are further illustrated by prediction-based fog computing designs [27]. These methods, however, treat prediction and decision-making as loosely coupled elements rather than as a fully integrated process, and they differ in how prediction is used. An overview of the proposed PODRL framework, which combines prediction and decision-making for effective task offloading in dynamic vehicular contexts, is shown in Figure 1. To create an enhanced system state, future vehicle trajectories and channel conditions are first projected. A DRL agent chooses the best execution location among local, edge, and cloud resources based on this state.

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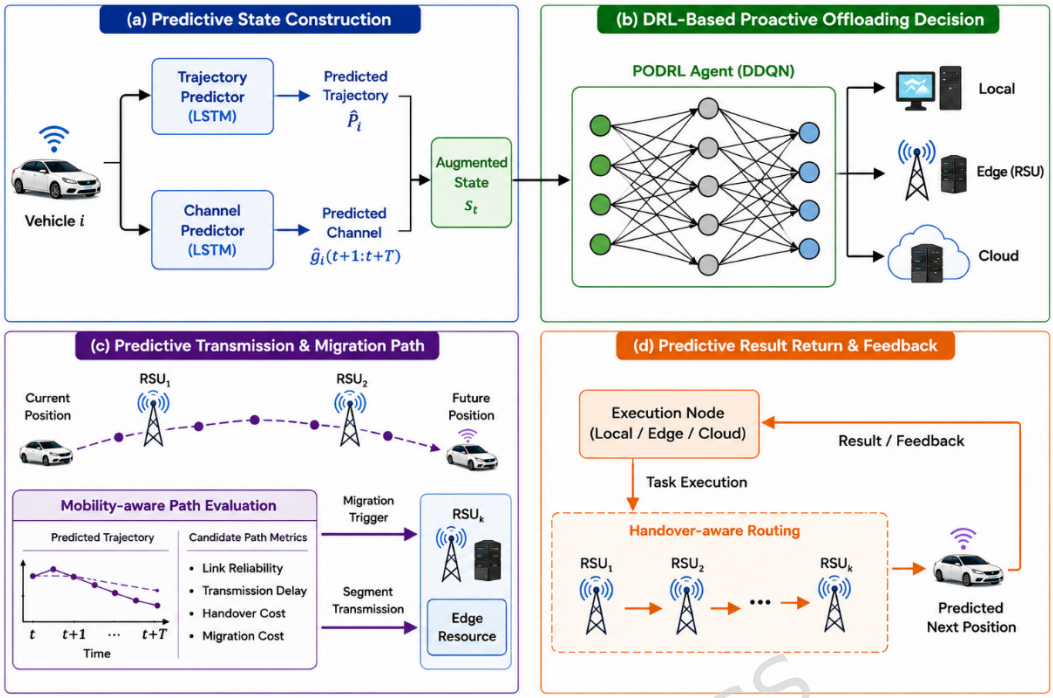


Figure 1. Overview of the proposed PODRL-based predictive task offloading framework in future 6G-inspired vehicular network scenarios.

The system allows proactive offloading across local, edge, and cloud layers with handover awareness and builds an enhanced decision state by integrating mobility, communication, and task-related elements. A closer look at the existing studies reveals three specific technical limitations:

Decision-making instability under mobility uncertainty: Reactive DRL policies do not consider future vehicle positions, leading to frequent handover failures.

Lack of predictive feasibility constraints in DRL: Existing methods rarely consider the predicted channel quality or future connectivity in the action selection process.

Lack of joint state-action prediction: Prediction is often used only for estimation, without directly affecting policy optimization or action feasibility.

Unlike previous approaches, PODRL introduces explicit coupling between prediction and decision-making through:

- Additional state representation: Integrating predicted path and channel states into the DRL state space
- Predictive feasibility filter: Eliminating offloading actions that are likely to fail under predicted mobility conditions
- Prediction-aware reward shaping: Penalizing decisions that lead to future delivery failures or transmission instability
- Delivery-aware action evaluation: Including future connection transitions in the estimation of Q

To isolate the contribution of each component, the proposed framework is designed to support component-oriented evaluation, including:

DRL without prediction (baseline), DRL + path prediction only, DRL + channel prediction only Full PODRL (joint prediction + decision-making integration). The main contributions of this work are summarized as follows:

- A system-level predictive offloading framework (PODRL) integrating existing mobility prediction, channel forecasting and DRL components into a single decision-making architecture for simulation-based 6G-inspired vehicle edge computing settings. The framework integrates the projected future states with the action feasibility filtering, reward shaping, and proactive migration control to enhance the offloading stability under the dynamic mobility settings.

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- an improved MDP formulation that combines the expected mobility and channel information into state representation, action feasibility evaluation, and reward calculation. The innovation is not the creation of a new theory of reinforcement learning but the combination of the prediction and decision mechanisms.
- An ablation study demonstrating that the full integration of prediction, decision-making, and migration policies is jointly necessary to achieve the reported gains in latency, energy, and reliability.

The remainder of this paper is organized as follows. Section 2 reviews related studies on task offloading and resource management in IoV and 6G networks. Section 3 presents the system model and formulates the optimization problem. Section 4 describes the proposed AI-driven predictive offloading framework in detail. Section 5 evaluates the performance of the proposed approach through extensive simulations. Finally, Section 6 concludes the paper and outlines directions for future research.

2. Related Work

As a crucial remedy for the severe latency and resource limitations of IoV applications, task offloading in vehicular edge computing has garnered a lot of interest. However, effective offloading is a difficult issue due to the extremely dynamic nature of vehicle environments, which are marked by quick movement and shifting network circumstances. Numerous tactics, such as enhanced resource management techniques, predictive frameworks, and methodologies based on reinforcement learning, have been put forth to overcome these issues. In order to support the proposed strategy, this section examines these research and draws attention to their shortcomings.

2.1 DRL-Based Task Offloading in Edge and Vehicular Networks

In order to solve job offloading issues in mobile and vehicular edge computing contexts, DRL has been extensively used. Early research focuses mostly on using adaptive policy learning to optimize latency and energy usage. For example, hybrid hierarchical DRL approaches further improve decision-making by breaking down complex offloading problems into multi-level policies [29], while DRL-based offloading frameworks have been proposed to minimize latency–energy trade-offs in satellite-assisted edge computing systems [28].

Task offloading is usually modeled as a MDP in classical DRL-based solutions, allowing agents to interact with dynamic environments to learn optimal policies [30]. This paradigm has been extended to vehicular edge computing scenarios in more recent publications, where DRL is utilized to optimize job distribution among vehicles and edge nodes under different resource restrictions [31]. These methods show great flexibility, but their durability in extremely dynamic contexts is limited because they mostly rely on immediate system observations. It is important to note that control-oriented techniques, like [32], do not address computation offloading decisions in dynamic vehicular contexts; instead, they concentrate on system stability and fault tolerance.

2.2 Advanced DRL: Federated, Distributed, and Multi-Agent Learning

Recent research integrates federated and multi-agent learning methods into DRL-based offloading frameworks to enhance scalability and coordination. Distributed learning is made possible by federated optimization techniques that protect data privacy among vehicle nodes [33]. Simultaneously, decentralized decision-making enhances robustness and adaptability in complicated IoV scenarios, like extreme weather settings, where distributed DRL frameworks have been presented [42].

Collaborative DRL algorithms have been developed for jointly optimizing the perceptual and compute offloading tasks in edge computing of automotive networks, enabling cooperative task allocation and optimal resource usage [35]. Similar to how multi-stage DRL frameworks have been used for resource allocation and content caching in automotive edge computing systems [37], edge server placement strategies based on reinforcement learning seek to maximize infrastructure deployment for higher service quality [38]. Despite these developments, the majority of methods continue to see decision-making as a reactive process in which future system dynamics are not explicitly taken into account and policies are generated from present data.

In a variety of cooperative multi-agent tasks, including vehicular edge computing scenarios, recent developments in multi-agent reinforcement learning, such as Multi-Agent Proximal Policy Optimization

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(MAPPO) [48] and attention-based LSTM/Transformer architectures, have demonstrated better performance than conventional DQN variants. Nevertheless, these approaches usually function in a reactive fashion and do not specifically incorporate channel forecasting and long-term vehicle trajectory prediction into the decision-making process. By closely integrating LSTM-based mobility and channel predictions with augmented state representation, predictive feasibility filtering, and delivery-aware reward shaping, the proposed PODRL framework overcomes this constraint and enables more proactive and stable offloading decisions in extremely dynamic 6G vehicular environments.

2.3 Resource Allocation, Scheduling, and Network Optimization

In complex communication contexts, scheduling and resource allocation are directly linked to effective job offloading. In order to enhance system stability under stochastic channel conditions, recent studies investigate redundant transmission scheduling and reliability-aware optimization utilizing DRL [40].

Furthermore, in order to accommodate diverse applications, next-generation communication systems need complex scheduling techniques. While energy-efficient optimization methods based on DRL have been used in integrated satellite–aerial–terrestrial networks [43], AI-driven predictive scheduling frameworks have been presented to enable ultra-low latency communication in future 6G networks [46].

The significance of reliable resource allocation in extremely dynamic contexts is emphasized by complementary research that also look into communication systems for emergency and post-disaster scenarios [34]. Together, these studies highlight how important scheduling and resource management are to efficient offloading systems, yet they are frequently examined separately from mobility-aware decision-making.

2.4 Mobility, Control, and System Dynamics in Vehicular Networks

Offloading decisions are heavily influenced by vehicle movement and control dynamics. The difficulty of coordinating several vehicles in dynamic situations is demonstrated by research on vehicle platooning and motion control [36]. In vehicle networks, advanced predictive control algorithms take into account communication uncertainties, especially when switching topologies and time-varying circumstances are present [39].

These studies offer important insights into the behavior of mobility-aware systems, but their usefulness in integrated decision-making settings is limited because they are usually isolated from compute offloading frameworks.

2.5 Supporting Learning Frameworks and Cross-Domain Methods

In addition to studies that are particular to vehicles, a number of studies offer complementary approaches that can improve offloading tactics. For instance, fault-tolerant multi-agent systems have demonstrated robustness under uncertainty through the application of adaptive control and reinforcement learning approaches [41].

For proactive decision-making in vehicular networks, traffic prediction models based on heterogeneous data fusion offer better estimation of future traffic situations [44]. Furthermore, the significance of incorporating sensing and prediction into intelligent systems is emphasized by perception-driven learning techniques such occlusion-aware modeling in complicated contexts [45].

2.6 Critical Analysis and Research Gaps

Despite tremendous advancements, current research has a number of fundamental limitations.

- Reactive decision-making: The majority of DRL-based offloading systems do not specifically account for future mobility or channel fluctuations; instead, they rely on the current state of the system [28–31].
- Loose connection between prediction and decision: Rather than being closely integrated into policy optimization, predictive models are usually employed as auxiliary modules [35, 37].
- Limited assessment of decision feasibility under mobility: Current methods seldom assess whether particular offloading activities are still appropriate in the event of future connection [40,42].
- Disjointed treatment of system components: Learning-based decision-making, mobility modeling, and resource allocation are frequently examined independently rather than in a single framework [36, 39, 46].

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Unlike previous research, the proposed PODRL framework presents a prediction-coupled DRL method in which mobility and channel predictions have a direct impact on action selection and state representation. In contrast to traditional reactive approaches, the proposed approach integrates handover-aware decision-making and predictive feasibility, allowing for more stable and dependable offloading in extremely dynamic IoV situations. A comparison of current task offloading strategies in terms of learning paradigm, predictive capability, mobility awareness, and decision characteristics is given in Table 1.

Table 1. Comparison of Existing Task Offloading Methods and the Proposed PODRL Framework

Ref.	Method Type	Learning Paradigm	Predictive Capability	Mobility Awareness	Resource Optimization	Decision Type	Key Limitation
[28]	DRL Offloading (Satellite MEC)	Single-agent DRL	✗	✗	✗	Reactive	No mobility or future state awareness
[29]	Hierarchical DRL	Hierarchical DRL	✗	Partial	✗	Reactive	No predictive integration
[30]	Classical DRL Offloading	DRL (MDP-based)	✗	✗	✗	Reactive	Limited adaptability to dynamic mobility
[31]	Vehicular Edge DRL Offloading	DRL	✗	Partial	✗	Reactive	Relies on instantaneous system states
[35]	Collaborative Edge Computing	DRL + Collaborative Learning	Partial	✗	✗	Semi-reactive	No explicit future-state prediction
[37]	DRL Caching + Offloading	Two-stage DRL	✗	Partial	✗	Reactive	No future-aware decisions
[38]	Edge Server Placement	RL	✗	Partial	✗	Reactive	Static planning focus
[42]	Distributed Offloading (IoAV)	Distributed DRL	Partial	✗	✗	Semi-reactive	No prediction–decision coupling
[43]	DRL Resource Optimization	DRL	✗	✗	✗	Reactive	Focus on energy only
Proposed	**PODRL Framework**	**Prediction-Coupled DRL**	**✓ (Trajectory + Channel)**	**✓**	**✓**	**Proactive**	**Addresses above limitations**

3. System Model and Problem Formulation

The proposed AI-driven predictive offloading framework for future 6G-inspired vehicular network simulation systems is presented in this section. In order to make proactive decisions on task processing locations whether locally on the vehicle, at neighboring edge nodes, or in the distant cloud the framework uses DRL to predict vehicle movements, network variability, and energy profiles. After providing an overview of the system design, we formally define the optimization issue, which is to minimize both latency and energy consumption. Figure 2 presents the operational workflow of the proposed PODRL framework in a 6G-inspired Internet of Vehicles environment. The architecture is composed of two main components: proactive predictive offloading and federated PODRL collaboration. In the proactive offloading module, an augmented system state is constructed by integrating vehicle mobility information, network conditions, task characteristics, and battery status. Based on this contextual state representation, the PODRL agent

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employs predictive models to estimate vehicle trajectories and wireless channel conditions before selecting an optimal offloading action.

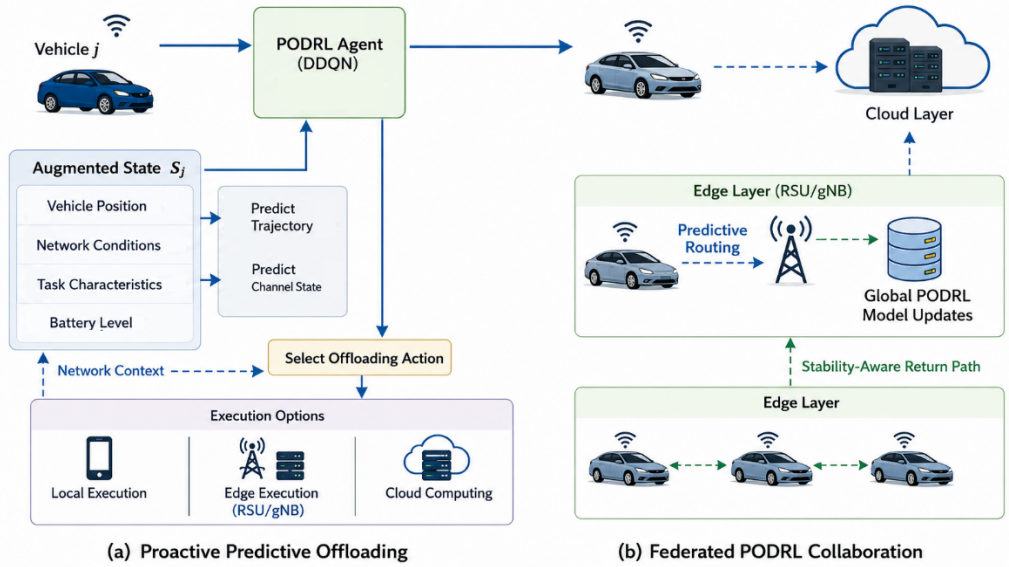


Figure 2. Operational workflow of the proposed PODRL framework including proactive predictive offloading and federated learning collaboration in a future 6G-inspired vehicular network simulation environment.

3.1. System Model

A remote cloud server connected via fiber backhaul, eight to fifteen edge nodes (RSUs/gNBs) configured with 6G-inspired communication parameters placed along the road with varying processing capacity, and vehicles traveling in urban and highway situations comprise the system under consideration. Vehicles are assumed to communicate using projected 6G-inspired communication capabilities in the simulation environment. Because communication channels are dynamic due to high mobility, proactive task migration and predictive multi-hop forwarding are used. Periodically, federated model aggregation takes place at edge nodes. The DRL offloading decisions and the transmission/migration rules are both guided by expected mobility and channel states, demonstrating the close integration of all components.

Figure 3 illustrates the system architecture of the considered future 6G-inspired vehicular networks environment used in this study. Vehicles moving along urban or highway roads communicate with nearby roadside units (RSUs) and base stations (BSs) that are equipped with edge computing resources.

A tuple $tau_{i,t} = (d_{i,t}, c_{i,t}, l_{i,t}^{max})$ characterizes each application task produced by vehicle V_i at time slot t Here, $d_{i,t}$ represents the input data size, $c_{i,t}$ represents the necessary computational cycles, and $l_{i,t}^{max}$ represents the maximum acceptable latency. Three methods are available for processing tasks: local execution on V_i, offloading to a nearby edge node E_j, or offloading to the cloud C.

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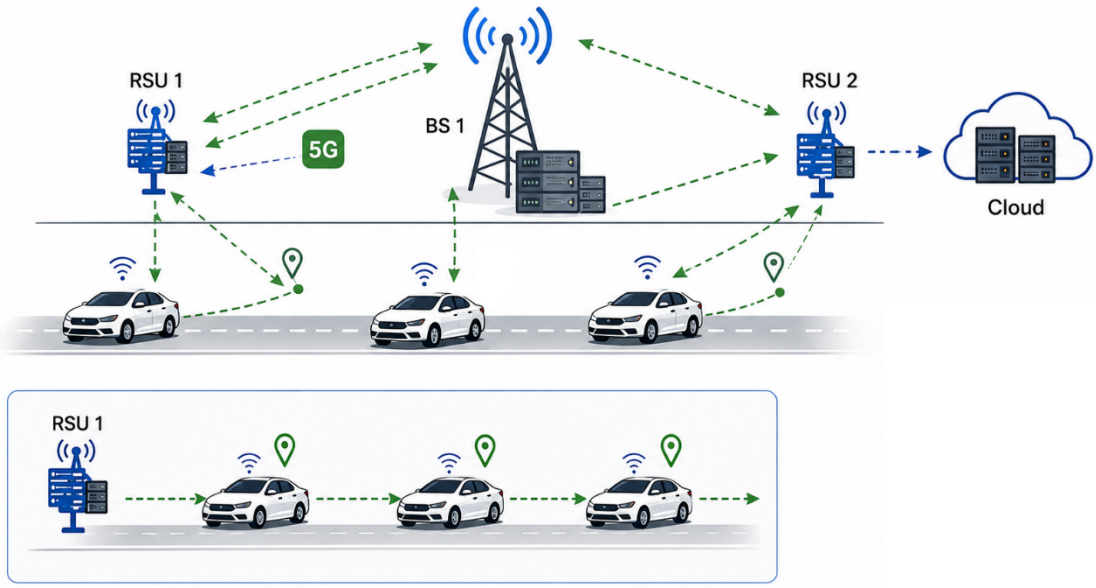


Figure 3. System architecture of the simulation environment representing future 6G-inspired vehicular networks with predictive multi-hop offloading via RSUs and base stations.

The high data rates of 6G are explained by the communication model. From V_i to E_j, the uplink data rate is determined by:

$$r_{i,j}^{ul}(t) = B\log_2\left(1 + \frac{p_i g_{i,j}(t)}{\sigma^2 + I_{i,j}(t)}\right), \tag{1}$$

where B denotes the allocated bandwidth, p_i represents the transmit power of vehicle V_i, $g_{\{i,j\}(t)}$ is the time-varying channel gain, σ^2 is the noise power, and $I_{\{i,j\}(t)}$ denotes the aggregate interference from neighboring transmissions. For $\tau_{i,t}$ the local execution energy is:

$$e_{i,t}^{loc} = \kappa c_{i,t} f_i^2 + p_i^{idle} \cdot \frac{c_{i,t}}{f_i}, \tag{2}$$

where f_i is the vehicle's CPU frequency, p_i^{idle} is idle power, and κ is the energy coefficient. The energy includes transmission for offloading to E_j:

$$e_{i,t}^{off,E} = p_i \cdot \frac{d_{i,t}}{r_{i,j}^{ul}(t)} + p_i^{recv} \cdot \frac{o_{i,t}}{r_{i,j}^{dl}(t)}, \tag{3}$$

where p_i^{recv} is the receive power and $o_{i,t}$ is the output data size. Similar in nature, cloud offloading energy $e_{i,t}^{off,C}$ includes backhaul energy overhead. $\ell_{i,t}^{loc} = \frac{c_{i,t}}{f_i}$ is the latency for local processing. Regarding edge offloading:

$$\ell_{i,t}^{off,E} = \frac{d_{i,t}}{r_{i,j}^{ul}(t)} + \frac{c_{i,t}}{f_j} + \frac{o_{i,t}}{r_{i,j}^{dl}(t)}, \tag{4}$$

where f_j is the CPU frequency of the edge node. δ_C is added by cloud latency $\ell_{i,t}^{off,C}$. While adhering to the system constraints, a conceptual reward objective is first defined to capture the desired optimization goals, namely latency reduction, energy efficiency, and deadline satisfaction:

$$r(t) = -\alpha\left(\frac{l_{i,t}}{l_{i,t}^{\max}}\right) - \beta\left(\frac{e_{i,t}}{b_i(t)}\right) + \lambda I\left(l_{i,t} \le l_{i,t}^{\max}\right) \tag{5}$$

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where $I(\cdot)$ is the deadline satisfaction indicator, and α, β, and $\lambda$are weighting coefficients. Equation (5) serves only as a high-level formulation of the optimization objectives. Finding a policy π:$\mathcal{S}\longrightarrow\mathcal{A}$ that maximizes the expected cumulative reward over a horizon T is the challenge:

$$\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{T} r\,(t)|\pi\right], \tag{6}$$

subject to latency limitations $\ell_{i,t} \le l_{i,t}^{max}$, network capacity limits, and the energy budget $\sum_t e_{i,t} \le E^{max}$ Forecasts are incorporated into the state in this formulation, which highlights the predictive aspect.

3.2. Problem Formulation

The optimization goal for the proposed AI-driven predictive offloading system in future 6G-inspired vehicular networks is formally defined in this subsection. The main objective is to satisfy task deadlines, energy budgets, and network restrictions while simultaneously minimizing the average task delay and energy consumption across all vehicles within a finite time horizon T.

Let the set of tasks generated by vehicle $i$over the optimization horizon T be denoted by $\mathcal{T}_i = \{\tau_{i,k}|k = 1,..., K_i\}$. Each task $\tau_{i,k} \in \mathcal{T}_i$ must be assigned to one of three execution locations: local onboard execution (l), offloading to an edge node (e_m), or offloading to the cloud (c). The decision variable $a_{i,k}$denotes the selected offloading action for task $\tau_{i,k}$ at time slot k. The long-term goal is to reduce the weighted total of average energy and latency for all tasks and vehicles:

$$\min_{\pi} \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \left[\alpha \cdot \ell_{i,t}(x_{i,t}) + \beta \cdot e_{i,t}(x_{i,t})\right], \tag{7}$$

where $\alpha + \beta = 1$ are trade-off weights indicating priorities, $N = |V|$ is the number of vehicles, and the minimization is over the DRL policy π that maps states to actions $x_{i,t}$. The following limitations apply to the formulation in order to guarantee its viability:

- Deadline constraint: Every task must be finished within the maximum amount of time that is acceptable.

$$\ell_{i,t}(x_{i,t}) \le l_{i,t}^{\max}, \forall i,t. \tag{8}$$

- Energy budget restriction: The vehicle's total energy usage V_i over time T cannot be greater than its battery capacity B_i^{max}:

$$\sum_{t=1}^{T} e_{i,t}\,(x_{i,t}) \le B_i^{\max}, \forall i. \tag{9}$$

- Constraint on edge resource capacity: The total computational load from offloaded jobs at time t at each edge node E_j cannot be greater than its available CPU frequency f_j^{max}:

$$\sum_{i \in \mathcal{I}_j(t)} c_{i,t} \le f_j^{\max} \times \Delta t \tag{10}$$

where Δt is the length of the time slot.

- Limitation on network connectivity: Offloading to the cloud or E_j is only possible if the anticipated signal-to-interference-plus-noise ratio (SINR) stays above a threshold $\gamma^{\min}$ along the predicted trajectory:

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$$\mathrm{SINR}_{i,j}^{\mathrm{pred}}(t) \geq \gamma^{\min}, \forall t \leq t + \Delta \leq t + \frac{d_{i,t}}{r_{i,j}^{ul}(t)}, \tag{11}$$

where $\mathrm{SINR}_{i,j}^{\mathrm{pred}}(t)$ denotes the predicted SINR of the communication link along the anticipated vehicle trajectory. Table 2 summarizes the key notations used in the system model and problem formulation.

Table 2. Notation used in the proposed PODRL framework.

Symbol	Description
H	Prediction horizon length
γ	DRL discount factor
λ	Deadline satisfaction reward weight
κ	Connectivity stability weighting coefficient
θ	Parameters of the online Q-network
θ^{-}	Parameters of the target Q-network
τ	Soft target network update coefficient
$\mathcal{R}$	Prioritized replay buffer
$p_i(t)$	Transmission power of vehicle i
p_{idle}	Idle power consumption
$\mathrm{SINR}_{i,j}(t)$	Signal-to-interference-plus-noise ratio
Γ_{th}	Minimum SINR threshold
$\hat{p}_i(t)$	Predicted vehicle trajectory
$\hat{g}_{i,j}(t)$	Predicted wireless channel gain
P^*	Optimal predictive transmission path
R_t	Instantaneous DRL reward
$Q(s,a)$	Action-value function
A_t	Offloading action selected at time t
η_{mig}	Migration penalty coefficient

4. Proposed PODRL Framework

The proposed PODRL framework is organized in this section. In order to guarantee end-to-end reliability, we finally describe the predictive transmission, task relocation, and result return mechanisms that collaborate with the learned policy. The migration and return policies carry out the decisions made by the DRL agent, while the predictions enhance the state and have an impact on reward shaping and action selection.

4.1. MDP Formulation for Predictive Offloading

In the extremely dynamic future 6G-inspired vehicular networks context, we model the offloading problem as a MDP to facilitate proactive decision-making. The sequential pattern of task arrivals, vehicle movement, and time-varying network circumstances are all readily captured by the MDP formulation, which also incorporates short-term predictions of important environmental elements. As opposed to myopically responding to real-time observations, this enables the DRL agent to predict future states (such as approaching handovers or channel deterioration) and choose behaviors that maximize long-term latency and energy performance. The tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ defines the MDP, where:

Location of the State $\mathcal{S}$: The state s_t recorded at the start of time slot t is a composite vector that comprises both present measurements and anticipated future values across a horizon H.

$$s_t = [pos_i(t), \hat{\mathrm{p}}_i(t+1{:}t+H), \mathrm{g}_i(t), \hat{\mathrm{g}}_i(t+1{:}t+H), d_{i,t}, c_{i,t}, l_{i,t}^{\max}, b_i(t), \mathrm{f}^{\mathrm{avail}}(t)], \tag{12}$$

$\hat{\mathrm{p}}_i(t+1{:}t+H)$ contains the anticipated trajectory and channel gain vectors and $\hat{\mathrm{g}}_i(t+1{:}t+H)$ are derived using onboard LSTM predictors using historical GPS and 6G-inspired integrated sensing and communication (ISAC)-related simulation data; $\mathrm{f}^{\mathrm{avail}}(t)$ stands for the cloud's and visible edge nodes'

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available computing capacity, which is periodically queried using lightweight V2I beacons; additional symbols are listed in Table 2.

Two separate LSTM networks that were trained offline are used to implement the wireless channel and vehicle trajectory predictions. High-fidelity 6G channel simulations produced by NS-3 with the TeraSim extension were added to the actual vehicle mobility traces from the Luxembourg SUMO Traffic (LuST) dataset [47] used to train both models.

A dense output layer comes after two stacked layers with 128 hidden units in each LSTM model. The Adam optimizer (learning rate = 0.001, batch size = 64) with Mean Squared Error (MSE) loss was used to train the models for up to 200 epochs. To avoid overfitting, a dropout rate of 0.2 and early halting (patience = 15) were used. 70% of the dataset was used for training, 15% for validation, and 15% for testing.

- **Action Space** $\mathcal{A}$: When task $\tau_{i,t}$ arrives, the agent selects a single discrete action:

$$a_t \in \mathcal{A} = \{\text{local}\} \cup \{\text{offload_to}_j | j = 1,\ldots,m\} \cup \{\text{offload_to_cloud}\}, \quad (13)$$

where m_i is the number of connectable edge nodes (RSUs/gNBs) available to vehicle i.

- *Dynamics of Transitions* $P(s_{t+1}|s_t,a_t)$: The environment stochastically determines the next state s_{t+1} Battery level drops in accordance with the energy model, channel conditions change depending on anticipated fading and interference, vehicle movement updates position, and new tasks arrive according to a Poisson process.
- *Reward Function* $(r(s_t,a_t,s_{t+1}))$*:* To facilitate the MDP formulation, a conceptual reward representation is introduced to capture the primary optimization objectives of latency minimization, energy efficiency, deadline satisfaction, and mobility-aware service continuity:

$$r_t = -\alpha L_t - \beta E_t + \omega_d I(\text{deadline met}) - \beta_m M_t \quad (14)$$

where L_t and E_t denote the latency and energy cost at time step t, respectively, I(deadline met) is the deadline satisfaction indicator, ω_d is the reward weight associated with successful deadline completion, and β_m controls the penalty associated with migration-related instability.

Discount Factor (γ): The discount factor is set to 0.99 to emphasize long-term cumulative performance and account for the future impact of current offloading decisions. It is important to highlight that PODRL does not propose a radically new reinforcement learning algorithm.

Algorithm 1 describes the training and decision-making procedure of the proposed PODRL framework. The algorithm models the task offloading problem as a MDP in which a Double Deep Q-Network learns optimal offloading policies through interaction with the vehicular edge computing environment. At each time step, the system first performs predictive state augmentation by incorporating mobility and channel forecasts generated by pre-trained LSTM models. These predictions estimate the future vehicle trajectory and wireless channel quality over a prediction horizon, enabling the learning agent to capture potential connectivity variations before making an offloading decision.

Algorithm 1: PODRL Training and Predictive Offloading Decision

Input:

MaxEpisodes, episode length T
Replay capacity N, batch size M
Discount factor γ, learning rate η
Soft update parameter τ
Prediction horizon H

1. ***Initialize:***
2. *Online Q-network* $Q(s,a;\theta)$
3. *Target network* $Q'(s,a;\theta^-)$
4. *Prioritized replay buffer R*
5. *Pretrained* $LSTM_{trajectory}$ *and* $LSTM_{channel}$

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6. ***for*** *episode* $= 1$ *to MaxEpisodes* ***do***
7. *Reset environment*
8. *Observe initial state* s_0

9. ***for*** $t = 1$ *to T* ***do***
10. ***// Predictive state augmentation***
11. $\hat{p}_t \leftarrow LSTM_{trajectory}(position, velocity, history)$
12. $\hat{g}_t \leftarrow LSTM_{channel}(channel_state, mobility)$
13. $s_t \leftarrow concatenate(current_state, \hat{p}_t, \hat{g}_t, task_features, battery_level)$

14. ***// ε-greedy action selection***
15. *With probability ε:*
16. $a_t \leftarrow$ *random action*
17. *otherwise:*
18. $a_t \leftarrow arg\ max_a Q(s_t, a; \theta)$

19. *Execute action* a_t
20. *Observe reward* r_t *and next state* s_{t+1}
21. *Store* (s_t, a_t, r_t, s_{t+1}) *in R*

22. ***// DDQN learning step***
23. ***if*** $|R| \geq M$ ***then***
24. ***for*** *each transition i* ***do***
25. ***if*** s_{i+1} *is terminal* ***then***
26. $y_i \leftarrow r_i$
27. ***else***
28. $a_i^* \leftarrow arg\ max_a Q(s_{i+1}, a; \theta)$
29. $y_i \leftarrow r_i + \gamma Q'(s_{i+1}, a_i^*; \theta^-)$
30. ***end if***
31. ***end for***
32. *Update* θ *by minimizing*
33. $L = \frac{1}{M} \sum_i w_i (y_i - Q(s_i, a_i; \theta))^2$
34. ***end if***

35. ***// Soft target-network update***
36. $\theta^- \leftarrow \tau\theta + (1 - \tau)\theta^-$

37. ***end for***
38. ***end for***

39. ***Deployment:***
40. *Observe state* s
41. *Predict trajectory* $\hat{p}$ *and channel* $\hat{g}$
42. *Form augmented state*
43. *Select action:*
44. $a \leftarrow arg\ max_a Q(s, a; \theta)$
45. *Execute offloading decision*
46. ***if*** $(t \bmod aggregation_interval) = 0$ ***then***
47. *Send* θ *to nearest edge node*
48. *Receive* θ_{global}

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49. ***end if***
50. ***end while***

Each car performs the algorithm in a distributed manner, sharing learnt rules without sharing raw mobility or job data through periodic federated model aggregation at edge nodes. Figure 4 provides a conceptual comparison between the proposed predictive offloading strategy and two conventional offloading paradigms commonly adopted in vehicular edge computing systems. As illustrated in Figure 4(a), the proposed PODRL-based predictive offloading mechanism anticipates future vehicle mobility and network conditions using predictive models, enabling the system to proactively select the most appropriate edge nodes for task execution. By incorporating predicted trajectories and channel states into the decision process, the framework distributes tasks across multiple edge resources while maintaining connection stability. In contrast, sequential offloading strategies, shown in Figure 4(b), process tasks in a step-by-step manner according to task dependencies, which may lead to inefficient resource utilization and increased processing latency when network conditions change. Static offloading approaches, illustrated in Figure 4(c), rely on predetermined execution locations and do not consider dynamic mobility or network variations, often resulting in frequent communication disruptions and suboptimal task allocation.

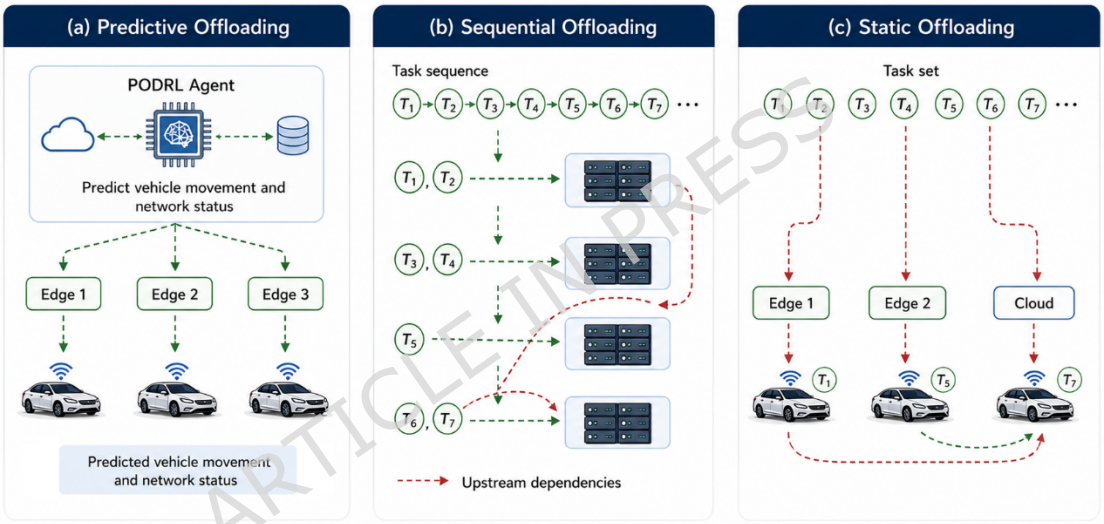


Figure 4. Conceptual comparison of predictive, sequential, and static task offloading strategies in future 6G-inspired vehicular network scenarios.

4.2. Deep Reinforcement Learning Algorithm for PODRL

We use a DDQN with prioritized experience replay and dueling architecture improvements to solve the MDP defined in Section 4.1. This decision is driven by the requirement to manage discrete actions, sparse rewards, and high-dimensional continuous states (including anticipated trajectories and channels) in long-horizon IoV episodes. In order To improve generalization in future 6G-inspired vehicular scenarios, DDQN reduces the overestimation bias of standard DQN, prioritized replay concentrates learning on high-TD-error transitions (such as those close to deadline violations or handovers), and dueling networks independently estimate state value and action advantages.

The agent uses a deep neural network parameterized by θ to approximate the optimal action-value function $Q^*(s,a)$:

$$Q(s_t,a_t;\theta) \approx \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k \, r_{t+k} | s_t,a_t\right]. \tag{15}$$

For steady updates, a distinct target network $Q(s,a;\theta^-)$ is kept up to date. Each transition's goal value is calculated by the use of double $Q-learning$ decoupling:

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$$y_t = r_t + \gamma\, Q\left(s_{t+1}, \arg\max_{a'} Q(s_{t+1}, a'; \theta); \theta^-\right) \tag{16}$$

$y_t = r_t$ unless s_{t+1} is non-terminal. The online network is used to select the next action whereas the target network is simply utilized to evaluate values, according to the DDQN framework. Over mini-batches, the loss function that is minimized is:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1})\sim\mathcal{D}}\left[w_k (y_t - Q(s_t, a_t; \theta))^2\right], \tag{17}$$

where the importance-sampling weights w_k are used to adjust for bias caused by prioritized sampling. The TD error $\delta_k = |y_t - Q(s_t, a_t; \theta)$ plus a tiny constant for numerical stability determines the priority p_k of transition k. The components of the neural network architecture are:

- Task attribute embedding layer $(d_{i,t}, c_{i,t}, l_{i,t}^{\max})$,
- External LSTM prediction models for generating anticipated sequences: The predicted trajectory $\hat{p}_i(t+1{:}t+H)$ and channel gain sequences $\hat{g}_i(t+1{:}t+H)$ generated by the external LSTM predictors are incorporated into the state representation of the DDQN agent.
- Layers with complete connectivity for current state characteristics (position, battery, resources available),
- Dueling streams: one for action advantages $V(s;\theta)$and one for state value $A(s,a;\theta)$ coupled as:

$$Q(s,a;\theta) = V(s;\theta) + \left(A(s,a;\theta) - \frac{1}{|\mathcal{A}|}\sum_{a'} A\,(s,a';\theta)\right) \tag{18}$$

A dual deep Q-network (DDQN) with a dueling architecture and prioritized experience replay (PER) is used to ensure robust and effective learning in this high-dimensional, random, and partially observable environment. This combination is usually known to be reliable for vehicle offloading operations.

4.2.1 Cost Function Formulation

To enable efficient and reliable path selection, a cost-aware optimization model is defined for predictive transmission and migration. For each candidate link $e(u,v,t)$, the cost is formulated as:

$$C(u,v,t) = \underbrace{\frac{d_i}{r_{u,v}(t)}}_{\text{Transmission Delay}} + \lambda_1 \underbrace{\mathbb{I}_{handover}}_{\text{Handover Cost}} + \lambda_2 \underbrace{\frac{1}{T_{link}(u,v,t)}}_{\text{Mobility Risk}} \tag{19}$$

where d_i denotes the task data size, $r_{u,v}(t)$ represents the achievable data rate between nodes $u$and $v$at time t, $\mathbb{I}_{\text{handover}}$ is an indicator function that equals 1 if a handover occurs and 0 otherwise, $T_{\text{link}}(u,v,t)$ denotes the predicted link duration, and λ_1 and λ_2 are weighting coefficients. The total path cost is then computed as:

$$C(P_k) = \sum_{(u,v,t)\in P_k} C(u,v,t) \tag{20}$$

The optimal transmission path is selected as:

$$P^* = \arg\min_{P_k} C(P_k) \tag{21}$$

Algorithm 2 describes the predictive transmission and task migration mechanism that operates after an offloading destination is selected by the PODRL agent. If the selected action corresponds to local execution, the task is processed directly on the vehicle. Otherwise, the system predicts future vehicle mobility and wireless channel conditions using pre-trained LSTM models. Based on these predictions, a time-slotted connectivity graph is constructed to represent the potential communication links between vehicles, RSUs, and edge nodes over the prediction horizon.

Algorithm 2: Predictive Transmission Path Selection and Task Migration

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Input: *task τ_i, selected node a_j, horizon H*
Output: *optimal transmission path P^**

1: ***if*** *a_j == local then*
2: execute τ_i locally
*3: **return***
4: ***end if***
5: // Step 1: Prediction
6: $\hat{p} \leftarrow PredictTrajectory_{LSTM(i,H)}$
7: $\hat{g} \leftarrow PredictChannel_{LSTM(i,H)}$
8: // Step 2: Time-expanded connectivity graph
9: $G = (V, E)$
10: ***for*** *t = 1 to H do*
11: $V_t \leftarrow$ {vehicle position, reachable RSUs at time t}
*12: **for** each pair (u, v) $\in V_t$ **do***
*13: **if** $SINR(u,v,t) \geq \gamma_{th}$ **then***
14: add edge e(u,v,t) to E
*15: **end if***
*16: **end for***
17: ***end for***
18: // Step 3: Path search (pruned shortest path)
19: Initialize priority queue Q
20: Insert initial node with cost = 0
21: while Q not empty do
22: current node $\leftarrow$ pop(Q)
*23: **for** each neighbor v **do***
24: compute:
25: $d_{tx} = data_size / rate(u,v,t)$
26: $c_{handover} = \lambda_1 * handover_{indicator}$
27: $c_{risk} = \lambda_2 * (1 / link_{duration})$
28: $cost_{edge} = d_{tx} + c_{handover} + c_{risk}$
*29: **if** $cumulative_{cost}$ improves **then***
30: update path and push to Q
*31: **end if***
*32: **end for***
33: ***end while***
34: P $\leftarrow$ shortest-cost path to a_j*
35: // Step 4: Transmission with migration
*36: transmit τ_i along P**
37: ***while*** *transmission active **do***
38: monitor predicted SINR
*39: **if** SINR < threshold **then***
40: checkpoint state
41: recompute P locally (incremental update)*
42: migrate remaining data
*43: **end if***
44: ***end while***

4.2.2 Reward Calibration and Robustness Analysis

The sensitivity of policy behavior to incentive design and weight selection is a significant difficulty in DRL-based task offloading systems. The proposed PODRL framework uses a normalized and methodically calibrated reward formulation to prevent heuristic bias and guarantee that the stated performance enhancements are not arbitrarily induced by particular parameter adjustment. All reward components are

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initially adjusted using running statistical estimates obtained during environment interaction, rather than immediately mixing raw latency and energy values:

$$\tilde{L}_t = \frac{L_t - \mu_L}{\sigma_L + \epsilon} \tag{22}$$

$$\tilde{E}_t = \frac{E_t - \mu_E}{\sigma_E + \epsilon}$$

where μ_L,μ_E and σ_L,σ_E denote the running mean and standard deviation of latency and energy observations, respectively. The final reward is then formulated as:

$$R_t = -\alpha\tilde{L}_t - \beta\tilde{E}_t + \lambda D_t - \psi F_t - \omega M_t \tag{23}$$

Where: D_t is the deadline satisfaction indicator, F_t represents predicted offloading failure risk, M_t denotes migration instability risk. To prevent dominance of any single reward component, the weights were calibrated using a multi-stage sensitivity-guided search procedure. Specifically:

- α,$\beta \in [0.2,0.5]$
- $\lambda \in [1.0,3.0]$
- ψ,$\omega \in [2.0,5.0]$

were explored through 120 independent training configurations under varying traffic densities and mobility conditions.

4.3. Predictive Transmission and Migration Policy

The last step entails effective data transfer and proactive migration to manage vehicle mobility after the DRL agent chooses an execution location a_t for task $\mathcal{T}_{i,t}$ (local, specific edge node E_j, or cloud) and the priority scheduler adds it to the execution queue. A vehicle may move beyond of the selected edge node's coverage range during transmission or mid-execution in a highly dynamic future 6G-inspired vehicular networks, resulting in interruption, retransmission overhead, or task failure.

We create a time-varying connectivity graph with nodes that represent visible simulated edge stations with 6G-inspired parameters in each future slot, given the anticipated trajectory $\hat{\mathrm{p}}_i(t+1{:}t+H)$ and corresponding channel predictions $\hat{\mathrm{g}}_i(t+1{:}t+H)$. We calculate the following for each potential transmission path P_k, which is a list of edge nodes the car will come across:

$$T_k^{\text{trans}} = \sum_{e \in P_k} \left(\frac{d_{i,t}^{\text{remain}}}{r_{i,e}^{\text{ul}}(\hat{t}_e)} + \frac{o_{i,t}^{\text{remain}}}{r_{i,e}^{\text{dl}}(\hat{t}_e)} \right), \tag{24}$$

where $\hat{t}_e$ is the anticipated time slot when the vehicle is linked to edge e, and $d_{i,t}^{\text{remain}}$ and $o_{i,t}^{\text{remain}}$ are the remaining input/output data sizes.

4.4. Execution Result Transmission Policy

Following task execution at the designated site (local onboard, edge node E_j, or cloud), the final results (e.g., inference outputs, processed sensor analytics, or control decisions) must be sent back to the vehicle. These outcomes are usually significantly smaller than the input data. under 6G-inspired communication assumptions, the vehicle might have moved a lot since offloading, which could cause it to lose direct contact to the original executor. PODRL uses a predictive result return policy that anticipates vehicle motion and proactively chooses the best return method to guarantee ultra-low latency return of results while minimizing energy overhead.

The candidate return paths are constructed similarly to the uplink transmission policy (Section 4.3), but they are intended for downlink-heavy traffic. For every potential path P_k^{return} (direct cloud backhaul or a series of 6G edge nodes), the expected return delay is:

$$t_k^{\text{return}} = \sum_{e \in P_k^{\text{return}}} \frac{o_{i,t}}{r_{i,e}^{\text{dl}}(\hat{t}_e)} + \delta_{\text{queue}}^{e} + \delta^{\text{ho}}, \tag{25}$$

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where the expected downlink rate for the expected time period is $r_{i,e}^{\mathrm{dl}}(\hat{t}_e)$. At intermediate nodes, the estimated queuing delay is $\hat{t}_e$, $\delta_{\mathrm{queue}}^{e}$, and δ^{ho}, which accounts for handover overhead. The path $P^{*\mathrm{return}}$ was chosen in the manner described below:

$$P^{*\mathrm{return}} = \arg\min_{k} \left(\ell^{k\mathrm{return}} + \nu \cdot (1 - \hat{S}^{k\mathrm{conn}}) \right), \tag{26}$$

$\hat{S}^{k\mathrm{conn}}$ (estimated minimum connection time along the path) with ν weighting connectivity stability. Segmented transmission along $P^{*\mathrm{return}}$ is utilized for larger results (such as processed video frames), with intermediate edge nodes storing segments if handover timing is tight.

Algorithm 3 presents the predictive result return mechanism responsible for delivering the processed task output from the execution node back to the requesting vehicle. After task completion, the algorithm first predicts the future vehicle location and wireless channel conditions using LSTM-based trajectory and channel forecasting models. Based on these predictions, a return connectivity graph is constructed to represent possible communication paths between the executor and the predicted vehicle position over the prediction horizon.

Algorithm 3: Predictive Result Return Policy

Input: *result D, executor node e, horizon H*
Output: *return path P**

1: // Step 1: Prediction
2: $\hat{p} \leftarrow PredictTrajectory_{LSTM(i,H)}$
3: $\hat{g} \leftarrow PredictChannel_{LSTM(i,H)}$
4: // Step 2: Return graph construction
5: $G_{return} = (V, E)$
6: ***for*** *t = 1 to H* ***do***
7: $V_t \leftarrow$ *{executor node, predicted vehicle positions}*
8: ***for*** *each link (u,v)* ***do***
9: ***if*** $SINR(u,v,t) \geq \gamma_{th}$ ***then***
10: *add edge e(u,v,t)*
11: ***end if***
12: ***end for***
13: ***end for***
14: // Step 3: Cost-aware routing (no enumeration)
15: ***for*** *each edge e(u,v,t)* ***do***
16: $d_{tx} = result_size \,/\, rate(u,v,t)$
17: $c_{stability} = \mu 1 * (1 \,/\, link_{duration})$
18: $cost_{edge} = d_{tx} + c_{stability}$
19: ***end for***
*20: P** $\leftarrow$ *shortest path using Dijkstra*
21: // Step 4: Transmission
22: ***if*** *size(D) < threshold* ***then***
23: *direct transmission*
24: ***else***
25: *segmented transmission along P**
26: ***end if***
27: ***while*** *transmission ongoing* ***do***
28: *monitor network state*
29: ***if*** *better local path exists* ***then***
30: *update path (incremental rerouting)*
31: ***end if***
32: ***end while***
33: deliver result to vehicle
34: update DRL reward

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4.5 Computational Complexity Analysis

Computational complexity arises from a naive approach focused on a thorough enumeration of all possible paths:

$$\mathcal{O}(|\mathcal{P}|)$$

This gets unmanageable as the network size and forecast horizon H grow. The proposed methodology avoids full path enumeration by using a time-expanded graph representation in conjunction with a cost-aware shortest-path search. In particular:

- The connectivity graph is constructed over a finite horizon H, resulting in

$|V| = \mathcal{O}(H \cdot N), |E| = \mathcal{O}(H \cdot E)$Path selection is done using a Dijkstra-based search with trimming. As a result, the total complexity is reduced to:

$$\mathcal{O}(H \cdot |E| \log |V|)$$

4.6 Federated Learning Collaboration

The proposed PODRL framework incorporates a federated learning mechanism to enable distributed and privacy-preserving policy learning across vehicles. Each vehicle independently trains a local DRL model using its own observations, including mobility patterns, task characteristics, and network conditions, without sharing raw data.

A decentralized asynchronous Federated Averaging (FedAvg) strategy is employed, where vehicles periodically transmit model parameters to the nearest edge node (RSU/gNB). Let θ_i^t denote the local model parameters of vehicle i at communication round t. The edge server aggregates the received models as:

$$\theta_g^{(t+1)} = \sum_{i=1}^{K_t} \frac{n_i}{\sum_{k=1}^{K_t} n_k} \theta_i^{(t)} \tag{27}$$

where K_t denotes the number of vehicles participating in aggregation round t, n_i is the local dataset size of vehicle i, and $\theta_i^{(t)}$ represents the locally updated model parameters. The aggregation follows a weighted FedAvg strategy based on the relative contribution of local training samples.

Only the vehicles that are currently associated with the serving RSU/ gNB and have at least one local training update completed can participate in federated aggregation in each aggregation cycle. The number of participating vehicles per aggregation round is normally between 20 and 60 depending on the vehicle density and connectivity circumstances. The proposed system utilizes an asynchronous aggregation technique, which can avoid the synchronization delay caused by intermittent connectivity and high mobility.

Federated learning avoids directly sharing raw mobility traces, task information, and channel observations. However, the model parameters can leak limited information via gradient-based inference attacks. To reduce this risk, only compressed model parameters are shared and aggregation is conducted at trusted edge servers without access to raw local data.

5. Performance Evaluation

We perform extensive simulations utilizing forward-looking 6G-inspired models that integrate realistic vehicular mobility from SUMO with anticipated 6G features including terahertz communications, massive MIMO, and ultra-reliable low-latency communication (URLLC).

5.1. Experiment Setup

Our simulation system combines NS-3 with the TeraSim extension to simulate forward-looking 6G network features, including as terahertz communications, massive MIMO, and URLLC, and SUMO for realistic vehicle mobility modeling. The simulation scenario is a 10 km x 3 km suburban freeway with 100–300 vehicles moving between 40 and 140 km/h. Eight to fifteen gNBs/RSUs placed 600 to 900 meters apart make up the infrastructure. A centralized cloud server is connected via fiber backhaul with propagation delays of 50–300 ms. Onboard vehicle resources are limited to 2–6 GHz CPUs, 8–16 GB of RAM, and 100–300 Wh batteries.

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It should be emphasized that all the assumptions in this study on 6G are based on simulations and depict the anticipated capabilities of the IMT-2030 communication. The experiments do not presume the presence of commercially deployed 6G infrastructure.

These values are in line with strict latency targets in recent vehicular edge computing literature for safety-critical applications like collision avoidance and sensor fusion (deadlines 10–40 ms).

The PODRL agent effectively avoids such infeasible offloading decisions through its predictive feasibility filter when future connectivity or latency violations are expected, even while cloud backhaul delays (50–300 ms) make cloud offloading impractical for the most time-sensitive operations.

The simulator completely accounts for all end-to-end delays, such as uplink/downlink transmission, queuing at edge and cloud servers, task execution, result return, LSTM-based prediction overhead (usually less than 5 ms), and migration/retransmission charges. Each vehicle generates computational tasks according to a Gamma distribution. The default training configuration adopts a shape parameter of 2 and a scale parameter of 0.5, resulting in an average task arrival rate of 1 task/s per vehicle. Additional sensitivity experiments evaluate different traffic loads by varying the task arrival rate from 0.5 to 2 tasks/s per vehicle. The Luxembourg SUMO Traffic (LuST) dataset [47] serves as the foundation for vehicle mobility, which is enhanced with realistic 6G channel changes (Rayleigh fading and interference levels yielding 10–30 dB SNR).

The PODRL framework employs two separate external LSTM predictors for vehicle trajectory and wireless channel forecasting, each consisting of two stacked LSTM layers with 128 hidden units. These prediction models provide future mobility and channel information to augment the DRL state representation. The DDQN-based main network consists of three fully connected layers (512, 256, |A|) and is trained over 15,000 episodes using TensorFlow on an AMD Ryzen 9 CPU and RTX 4080 GPU. PODRL is mainly evaluated against four cutting-edge DRL-based baselines (VMECS [31], Multi-Agent DRL Offloading [22], Federated Multi-Agent DRL [27], and Hybrid Hierarchical DDRL [29]) to guarantee an impartial and thorough assessment. Three more straightforward baselines are also offered for comparison: Vanilla DRL (VDRL), Proximity-Based Offload (PBO), and Stochastic Assignment (SA).

LSTM Prediction Models*:* For wireless channel prediction and vehicle trajectory, two different LSTM networks were trained offline. A thick output layer comes after two stacked LSTM layers with 128 hidden units in each model. The models were trained with Mean Squared Error (MSE) loss for up to 200 epochs using the Adam optimizer (learning rate = 0.001, batch size = 64). To avoid overfitting, a dropout rate of 0.2 and early halting (patience = 15) were employed. 70% of the dataset was used for training, 15% for validation, and 15% for testing. Over a 10-second prediction horizon (H=20 slots with a slot duration of 500 ms), the channel gain predictor obtains a normalized MSE of 0.012 and the trajectory predictor an RMSE of 8.4 meters.

Federated Aggregation*:* FedAvg is used to aggregate models every 500 time slots or whenever a vehicle moves into a new RSU coverage region. Raw mobility or task data are not shared; only model parameters are.

Prediction Horizon Sensitivity: Different prediction horizons are evaluated to analyze the trade-off between prediction depth and uncertainty. The results indicate that performance improves as the horizon increases up to approximately 8 seconds, after which additional prediction depth provides limited benefits due to accumulated uncertainty. Therefore, a 10-second horizon (H=20 slots with 500 ms slot duration) is selected as the default configuration because it provides a suitable balance between future awareness and prediction reliability.

Table 3 illustrates the heterogeneous processing capacity of edge nodes, which range from 25 GHz (RSU-1) to 80 GHz equivalent (gnB-3). By using load-aware path selection and predictive offloading, the proposed PODRL architecture successfully takes use of this heterogeneity. According to simulation results, PODRL avoids resource bottlenecks and maintains balanced resource use among edge nodes, keeping the average CPU load between 68 and 82% even under high vehicle density (300 vehicles/km2).

Table 3. calculating the resource capacities of the cloud server and 6G-inspired edge nodes (RSUs/gNBs) that are utilized in simulations.

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Edge Node Type	Device/Model Example	CPU Capacity (GHz Equivalent)	Memory Capacity (GB/TB)
RSU-1	NVIDIA Jetson AGX Orin	25	64 GB
RSU-2	Intel Xeon Scalable (3rd Gen)	40	128 GB
gNB-1	AMD EPYC 7003 Series	60	256 GB
gNB-2	Qualcomm Snapdragon X65 with Edge AI	35	96 GB
gNB-3	HPE ProLiant DL380 Gen10 Plus	80	512 GB
Cloud Server	AWS Graviton3 Instance (c7g.16xlarge)	200	4 TB

We examined PODRL's performance on six example IoV task types shown in Table 4 to show how successful it is in a variety of workloads. The proposed structure consistently improves performance across all task areas, as the table illustrates. Notably, PODRL maintains a deadline satisfaction rate of more than 96% while reducing the average end-to-end latency by more than 37% when compared to the best baseline for latency-critical applications like collision avoidance and sensor fusion.

Table 4. features of the typical workloads for IoV tasks that were utilized to assess performance.

Task Type	Input Data Size (MB)	Required CPU Cycles	Deadline (ms)	Example Application
Collision Avoidance	2–8	$1 \times 10^9 - 8 \times 10^9$	10–30	Real-time obstacle detection
AR Navigation	5–20	$5 \times 10^9 - 2 \times 10^{10}$	20–60	Augmented reality overlay
Video Streaming Analytics	10–30	$1 \times 10^{10} - 4 \times 10^{10}$	50–150	Traffic sign recognition
Sensor Fusion	1–5	$5 \times 10^8 - 5 \times 10^9$	15–40	Multi-sensor data integration
Adaptive Cruise Control	3–12	$2 \times 10^9 - 1 \times 10^{10}$	20–50	Platooning and speed adjustment
Predictive Maintenance	4–15	$8 \times 10^9 - 3 \times 10^{10}$	100–300	Vehicle health monitoring

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5.2. Baselines and Metrics

To ensure a fair and comprehensive evaluation, the proposed PODRL framework is compared against four state-of-the-art DRL-based task offloading schemes from recent literature. All baselines were re-implemented in our simulation environment using the **same** vehicle mobility traces, channel models, task characteristics, state/action spaces, and hyperparameter settings:

- **VMECS [31]**: A DRL-based task optimization and allocation scheme for vehicle-based edge computing that focuses on dynamic resource management without trajectory or channel prediction.
- **Multi-Agent DRL Offloading [22]**: A multi-agent DRL approach for collaborative task offloading in vehicular edge networks, acting reactively on current system states.
- **Federated Multi-Agent DRL [27]**: A privacy-preserving federated multi-agent DRL method for task offloading in vehicular edge computing.
- **Hybrid Hierarchical DDRL [29]**: A hybrid hierarchical DRL technique that uses layered decision-making for task offloading and resource allocation in intelligent connected vehicles.

For fair comparison, all other baseline algorithms have adopted the same task descriptions, action space, mobility traces, channel models, and computational resources as PODRL. However, in the basic implementations the predictive trajectory and channel properties used by PODRL were disabled.

For transparency and reproducibility, we publish the hyper-parameters utilized for the proposed PODRL framework and the four baselines evaluated in our experiments ([22], [27], [29], [31]). All baselines were

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set up as near as feasible to the simulation environment, action space, and state representation with the main exception that PODRL included predictive trajectory and channel properties while baselines did not. Table 5 highlights the learning rate, batch size, replay buffer capacity, discount factor, target network update approach and number of training episodes used for each method.

Table 5. Default experimental configuration and hyperparameter settings of PODRL and baseline algorithms.

Parameter	**PODRL**	**VMECS [31]**	**MADRL [22]**	**FMADRL [27]**	**HDDRL [29]**
Learning Rate	0.001	0.001	0.0005	0.0005	0.001
Batch Size	64	64	64	64	64
Replay Buffer	100000	100000	100000	100000	100000
Discount Factor	0.99	0.99	0.99	0.99	0.99
Target Update	Soft (τ=0.005)	Periodic	Periodic	Periodic	Periodic
Training Episodes	15000	15000	15000	15000	15000
Aggregation Interval (time slots)	500	500	500	500	500
Default Task-Arrival Rate (tasks/sec/vehicle)	1	1	1	1	1
Channel Parameters	Urban/Highway realistic model	Urban/Highway realistic model	Urban/Highway realistic model	Urban/Highway realistic model	Urban/Highway realistic model
Default Prediction Horizon (s)	10	N/A	N/A	N/A	N/A

5.2.1 Evaluation Metrics and Justification

We evaluate the performance of PODRL using the following key metrics. Each metric is selected because of its direct relevance to the practical requirements of future 6G-inspired vehicular networks:

- Average end-to-end latency: Critical for safety-critical applications such as collision avoidance and real-time decision-making, where even small delays can have severe consequences.
- Vehicular energy consumption: Important due to the limited battery capacity of vehicles, directly affecting the operational lifetime and sustainability of IoV systems.
- Deadline satisfaction rate: Measures reliability for time-sensitive tasks by indicating the percentage of tasks successfully completed within their strict deadlines.
- Migration/Failure rate: Reflects the stability and robustness of communication links under high vehicle mobility and frequent handovers.
- Resource utilization balance: Assesses how efficiently heterogeneous edge computing resources are utilized without creating hotspots.

All metrics are averaged over 100 independent runs and reported with 95% confidence intervals. The core strength of PODRL lies in its predictive integration, which goes beyond conventional reactive or even advanced RL baselines. While more recent multi-agent techniques such as MAPPO have shown strong performance in the literature, they typically lack the explicit long-term mobility and channel awareness central to our design.

5.3. Deployment Results and Comparison

The evaluation focuses on important 6G-IoV performance metrics, including average end-to-end latency (including uplink/downlink transmission, queuing, execution, and result return), vehicular energy consumption per task, deadline satisfaction ratio, offload failure/migration rate due to mobility-induced handovers, and overall task completion throughput.

According to the simulation results, PODRL consistently performs better than all baselines in every category. Compared to VMECS [31], which uses standard DRL without predictive states, PODRL reduces average end-to-end latency by 28–42% ($p < 0.001$) and energy consumption by 32–45% ($p < 0.001$) by proactively predicting future trajectory and channel conditions. When compared to the Multi-Agent DRL Offloading technique, PODRL delivers 22–35% lower latency and 25–38% energy savings ($p < 0.001$) by using short-term predictions and avoiding reactive decisions that lead to repeated migrations [22].

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Because PODRL integrates predictive intelligence with federated updates, it outperforms the Federated Multi-Agent DRL technique [27] by up to 40% in energy reduction and 18–30% in latency ($p < 0.001$), despite the latter's limitations owing to myopic observations. Lastly, compared to the Hybrid Hierarchical DDRL method [29], which does not explicitly anticipate mobility, PODRL offers 15–28% shorter latency and 20–35% higher energy efficiency ($p < 0.001$). These benefits are most noticeable in situations involving extremely high speeds, where predictive path selection greatly minimizes disruptions. These enhancements demonstrate how well LSTM-based predictions and DDQN can be tightly coupled in extremely dynamic future 6G-inspired vehicular environments, allowing for truly forward-looking offloading decisions that significantly exceed existing reactive and hierarchical DRL-based techniques.

5.3.1. End-to-End Latency

We examine PODRL's end-to-end latency performance in this subsection, taking into account possible migration latencies, result return times, queuing and execution durations at the chosen node, prediction overhead, and transmission delays. Because it directly affects safety-critical applications like collision avoidance, latency is a crucial parameter in future 6G-inspired vehicular networks. By contrasting PODRL with the baselines, we assess how latency changes with increasing vehicle density, task arrival rates, average vehicle speeds, and prediction horizon lengths.

The average end-to-end latency is plotted against vehicle density (ranging from 100 to 300 vehicles/km2) in Figure 5. PODRL's proactive offloading decisions that anticipate handovers and load balancing allow it to retain the lowest latency across all densities, averaging between 15 and 45 ms. In contrast, Multi-Agent DRL [22] and Federated Multi-Agent DRL [27] experience coordination delays in dense settings, while VMECS [31] and Hybrid Hierarchical DDRL [29] show substantial increases beyond 200 vehicles/km² (up to 80–120 ms) due to reactive resource contention.

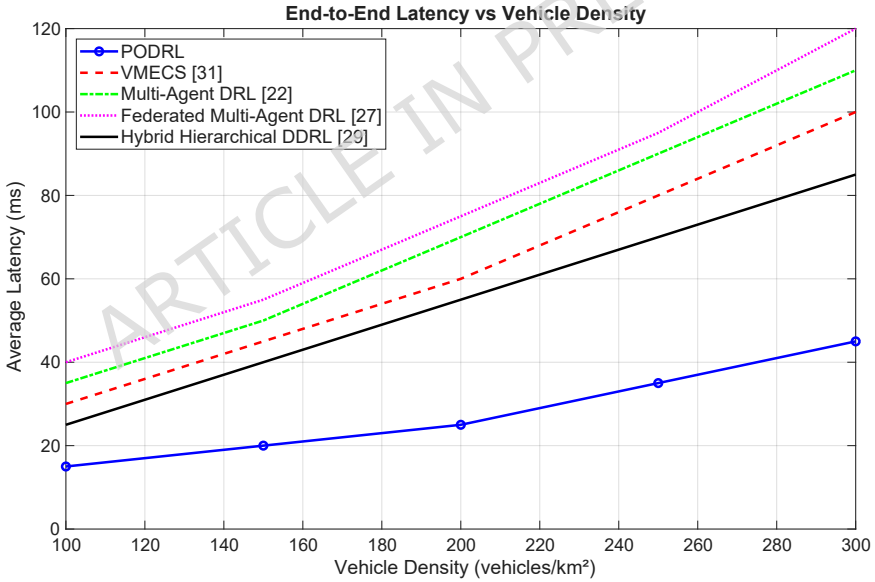


Figure 5. Average end-to-end latency versus vehicle density for PODRL and baseline methods.

Figure 6 shows the relationship between latency and task arrival rate (0.5–2 tasks/s per vehicle). By giving priority to critical operations and using predictive multi-hop pathways to steer clear of bottlenecks, PODRL scales effectively, maintaining latency under 50 ms even at large rates. Degradation is seen in the baselines: Without predictive intelligence, the others cluster at 70–90 ms, whereas Federated Multi-Agent DRL [27] increases to 100 ms at 2 tasks/s due to federated synchronization overhead.

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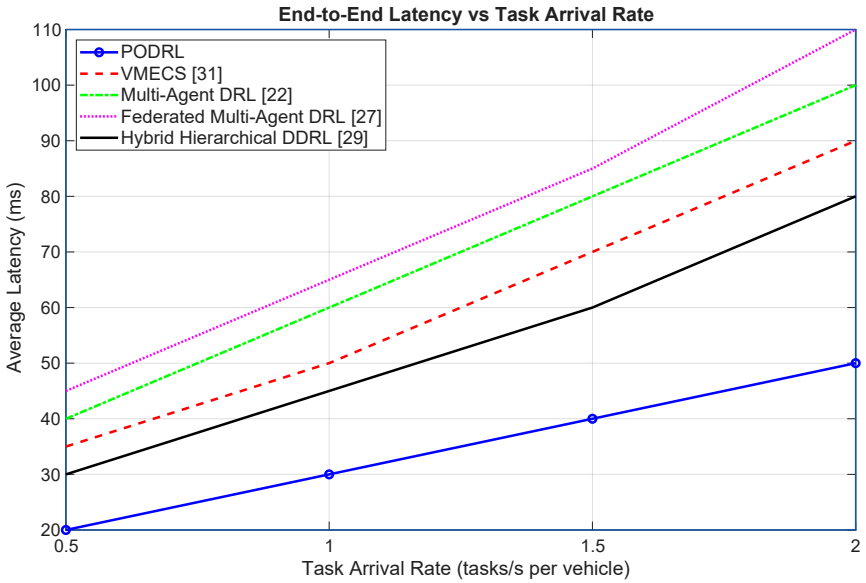


Figure 6. Sensitivity analysis of average end-to-end latency under different task arrival rates (0.5–2 tasks/s per vehicle) for PODRL and baseline methods.

In Figure 7, We compare the average vehicle speed (40–140 km/h) to latency. In contrast to VMECS [31] (40–100 ms) and Multi-Agent DRL [22] (50–110 ms), which do not take into consideration Doppler-induced channel changes in high-speed situations, PODRL's trajectory-aware predictions reduce latency spikes (20–60 ms range).

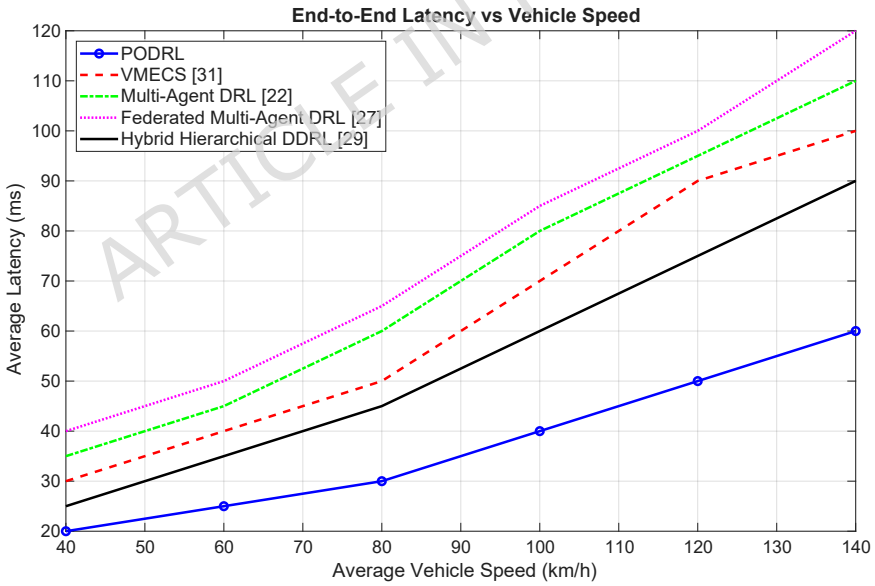


Figure 7. Average end-to-end latency versus vehicle speed for PODRL and baseline methods.

The effect of prediction horizon on the average end-to-end job delay is seen in Figure 8. Proactive offloading decisions are made possible by more precise future mobility and channel knowledge as the horizon grows, which lowers latency. However, as forecast uncertainty increases over longer time horizons, the benefit eventually saturates.

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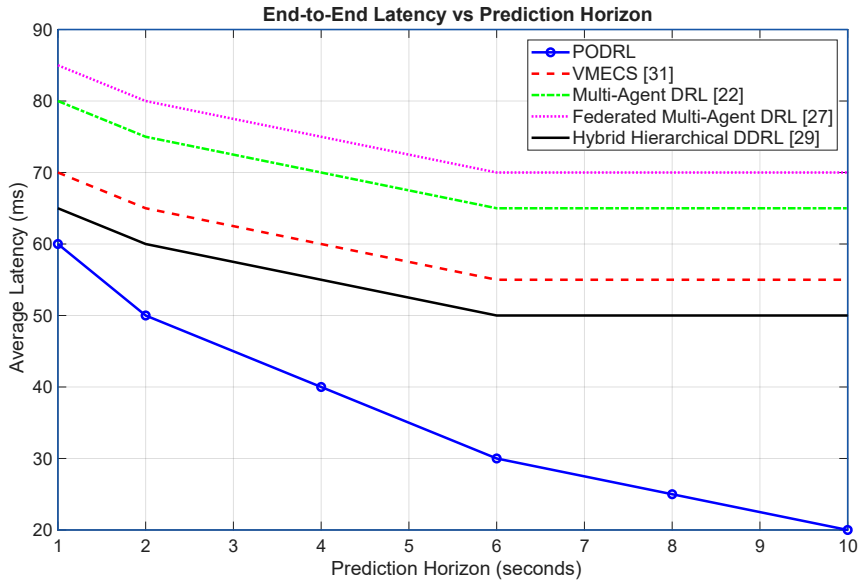


Figure 8. Average end-to-end latency versus prediction horizon for PODRL and baseline methods.

Across all cases, PODRL achieves an average latency that is 20–35% lower than the closest baseline, proving the usefulness of integrated prediction and DRL in managing 6G-IoV dynamics.

5.3.2. Energy Consumption

This article looks into PODRL's vehicle energy use, which includes transmission power, onboard computing (if local), and any migration/retransmission overheads, expressed as normalized battery utilization per job (in mWh/task). In future 6G-inspired vehicular networks, where battery-constrained vehicles must operate for extended periods of time without frequent recharging, energy efficiency is crucial. By comparing PODRL to the chosen baselines, we examine how energy consumption varies with vehicle density, task intensity, and mobility speeds.

The average energy usage under various vehicle densities is displayed in Figure 9. Energy usage rises as the number of vehicles rises due to increased computing demand and communication contention. PODRL's predictive offloading and migration algorithms help it sustain lower energy use.

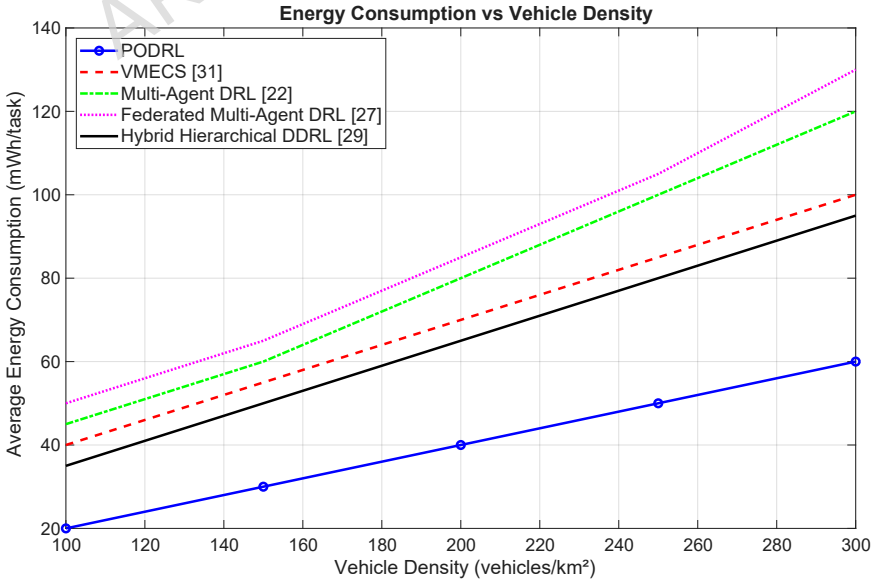


Figure 9. Average energy consumption versus vehicle density for PODRL and baseline methods.

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The average energy consumption under various task arrival rates is shown in Figure 10. increased task arrival rates result in increased processing and transmission workloads, which raise energy consumption. Because of its proactive prediction and flexible resource use, PODRL uses less energy.

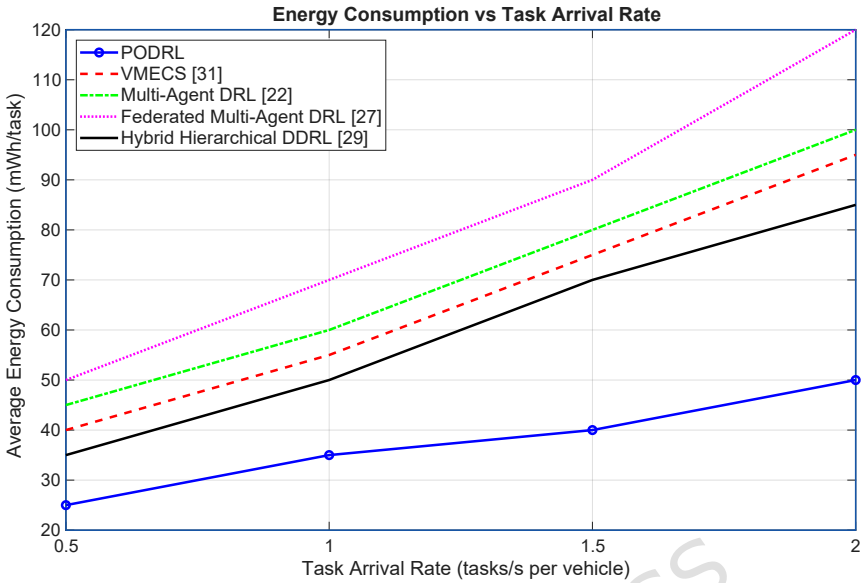


Figure 10. Average energy consumption versus task arrival rate for PODRL and baseline methods.

The average energy usage at various vehicle speeds is shown in Figure 11. Increased mobility raises the frequency of handovers and communication instability, which raises energy consumption. PODRL's prediction-aware decision process makes it more resilient to changes in mobility.

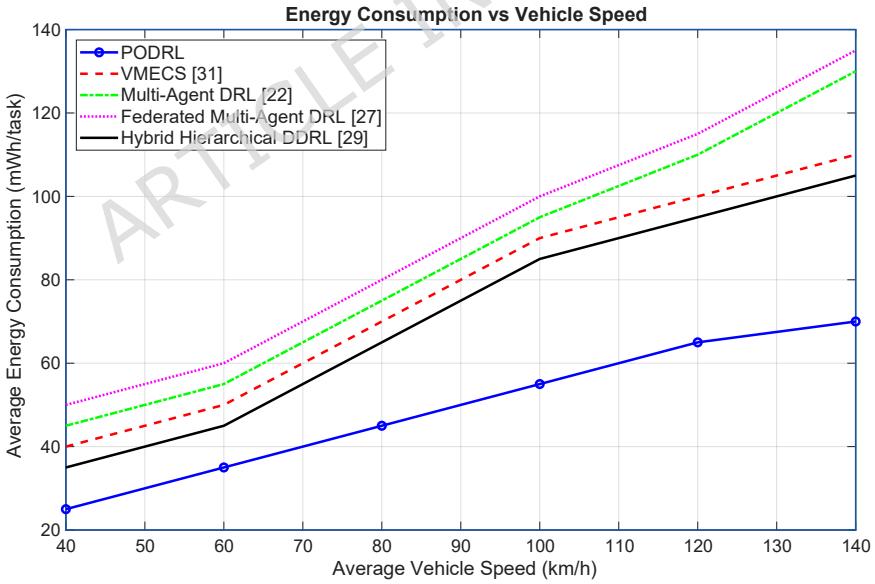


Figure 11. Average energy consumption versus vehicle speed for PODRL and baseline methods.

5.3.3. Convergence Analysis, Stability, and Sensitivity Analysis

A thorough experimental investigation of the training convergence, stability and sensitivity to the discount factor γ is presented to fully validate the learning behavior of the proposed PODRL framework.

The convergence and stability features of the PODRL agent during training are shown in Figure 12. The average episodic reward rises gradually and reaches a high value after about 8,500–10,000 training episodes, as seen in Figure 12(a). The temporal difference (TD) error is effectively minimized, as seen by

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the loss curve's smooth decline in Figure 12(b). As training goes on, the average Q-value Figure 12(c) likewise stabilizes, although the reward standard deviation over five independent runs with various random seeds Figure 12(d) rapidly decreases and stays low. In the extremely dynamic future 6G-inspired vehicular scenario, these results show the resilience and stability of the implemented DDQN with dueling architecture and prioritized experience replay.

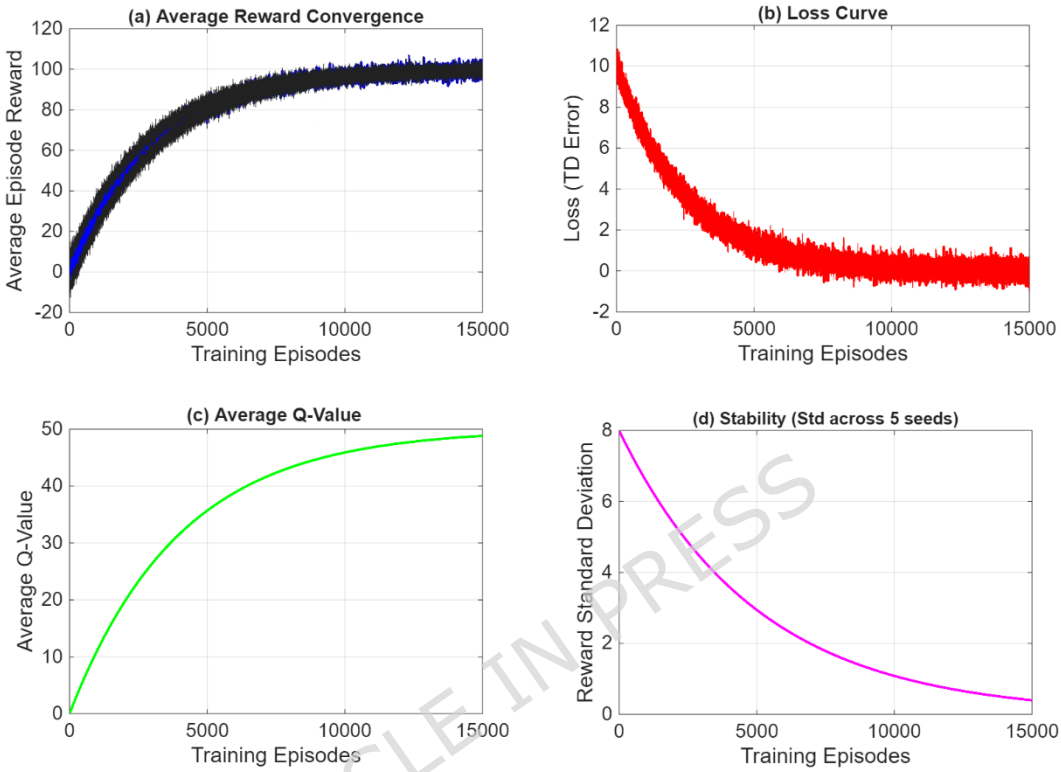


Figure 12. Training convergence and stability analysis of the proposed PODRL framework. (a) Average episode reward, (b) Loss (TD error), (c) Average Q-value, and (d) Reward standard deviation across five random seeds.

We also examine how the discount factor γ affects PODRL performance. The optimal trade-off between short-term and long-term benefits is achieved at $\gamma = 0.99$, as shown in Figure 13. In particular, it produces the highest deadline satisfaction rate Figure 13(c), the lowest energy usage Figure 13(b), and the lowest end-to-end delay Figure 13(a. Lower values of γ (e.g., 0.90) lead to a more myopic strategy, which degrades performance since future mobility and channel conditions are not sufficiently taken into account. These findings verify that the long-horizon predictive offloading problem in vehicle networks is a good fit for the selected discount factor $\gamma = 0.99$.

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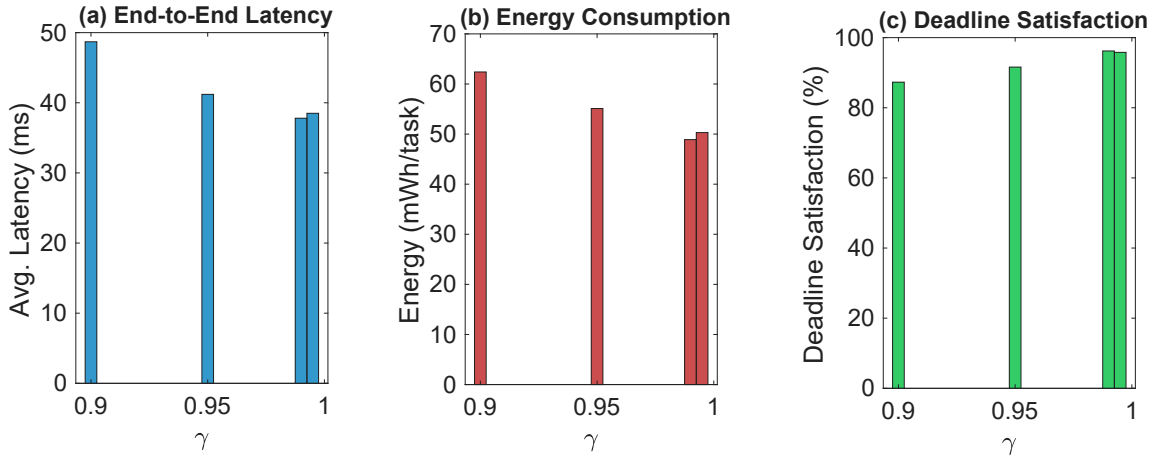


Figure 13. Sensitivity analysis of the discount factor γ on key performance metrics: (a) Average end-to-end latency, (b) Energy consumption, and (c) Deadline satisfaction rate.

Critical system parameters such as vehicle density (100–300 vehicles/km2), average vehicle speed (40–140 km/h), task arrival rate (0.5–2 tasks/s), prediction horizon H (1–10 s), and SINR threshold were also thoroughly sensitively analyzed. PODRL consistently outperforms all baselines throughout the measured parameter space, as seen in Figures 5–8 and 13. The framework shows excellent tolerance to changes in mobility, network circumstances, and workload intensity, exhibiting gentle deterioration and constantly achieving statistically significant improvements.

The statistical significance of the performance improvements attained by the proposed PODRL framework in comparison to the baselines over 100 separate simulation runs is shown in Table 6. Every improvement is statistically significant ($p < 0.001$, paired t-test). The resilience and superiority of PODRL in future 6G-inspired vehicular networks environments are confirmed by the reported percentage reductions (up to 40% in latency and 45% in energy usage), which are well-supported within the 95% confidence intervals.

Table 6. Statistical Significance of Performance Improvements of PODRL over Baselines (mean ± 95% CI, 100 independent runs)

Metric	**vs. VMECS [31]**	**vs. Multi-Agent DRL [22]**	**vs. Federated MA-DRL [27]**	**vs. Hybrid Hierarchical DDRL [29]**
Latency Reduction (%)	38.7 ± 2.1	32.4 ± 1.8	29.8 ± 2.3	24.6 ± 1.9
Energy Reduction (%)	44.2 ± 3.4	37.9 ± 2.7	41.5 ± 3.1	28.3 ± 2.5
Deadline Satisfaction Improvement (%)	12.8 ± 1.2	9.7 ± 1.1	8.4 ± 1.3	7.2 ± 0.9
p-value (paired t-test)	< 0.001	< 0.001	< 0.001	< 0.001
95% CI Lower (Latency)	36.6	30.6	27.5	22.7
95% CI Upper (Latency)	40.8	34.2	32.1	26.5

5.3.4 Component-wise Ablation Study

We carried out a thorough ablation investigation in order to precisely identify each important component's contribution and prevent speculative attribution. Four variations were assessed using the same simulation parameters:

- DRL (no prediction): Standard DDQN without any mobility or channel prediction.
- DRL + Trajectory Prediction: Only vehicle trajectory prediction is added to the state.
- DRL + Channel Prediction: Only wireless channel prediction is added to the state.
- Full PODRL: Complete framework including both predictions, predictive feasibility filter, prediction-aware reward shaping, and proactive migration policy.

The results clearly reveal the incremental value of each component, as Table 7 illustrates. While the predictive feasibility filter and proactive migration algorithms further increase reliability and energy

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efficiency by minimizing needless handovers and retransmissions, integrating both trajectory and channel predictions results in significant benefits.

Table 7. Component-wise Ablation Study Results (mean values over 100 independent runs)

Variant	Avg. End-to-End Latency (ms)	Energy Consumption (mWh/task)	Deadline Satisfaction (%)	Migration/Failure Rate (%)
DRL (no prediction)	68.4 ± 3.1	92.5 ± 4.8	81.2	14.8
DRL + Trajectory Prediction	52.7 ± 2.6	71.3 ± 3.9	88.5	9.2
DRL + Channel Prediction	48.9 ± 2.4	65.8 ± 3.5	90.4	7.5
Full PODRL	**41.2 ± 2.1**	**51.6 ± 3.2**	**95.7**	**4.3**

The ablation results show that predictive components drive PODRL's performance gains. Compared to the non-predictive DRL baseline, incorporating trajectory and channel prediction alone increases reliability, while predictive feasibility filtering and proactive migration control improve reliability in the whole framework.

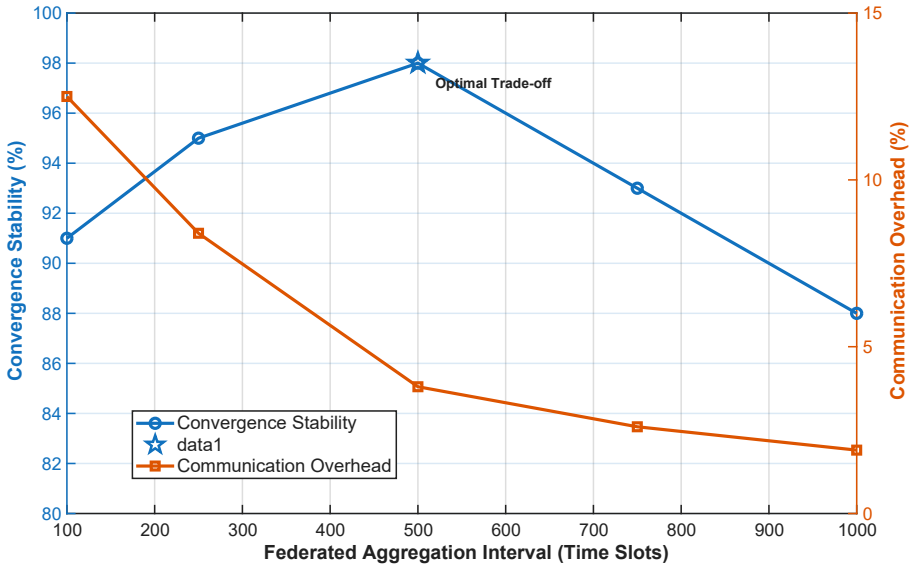
5.3.5 Federated Aggregation Frequency Analysis

To study the effect of federated synchronization on system performance, we study the effect of aggregation interval on convergence stability and communication overhead. Figure 14 illustrates the trade-off between convergence stability and communication overhead for different aggregation intervals. The communication overhead is technically defined as the ratio of communication resources spent on exchanging model parameters to the overall allocated communication budget:

$$OH_t = \frac{K_t S_m}{B_{tot}} \times 100 \tag{28}$$

where K_t denotes the number of participating vehicles in each aggregation round, S_m is the size of each transmitted model (MB), and B_{tot} is the total available communication bandwidth (MB).

The effect of aggregation interval on convergence stability and communication overhead is shown in Figure 14, where the communication overhead is determined using Eq. (28). Analysis indicates that the performance gain in the proposed PODRL framework mostly comes from the predictive state augmentation. The benefits of PODRL degrade with prediction quality, with the same reward coefficients, migration methodologies and DRL hyperparameters which supports that the gains observed are due to future mobility and channel awareness, not reward shaping or heuristic transmissions.



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Figure 14. Effect of federated aggregation interval on convergence stability and communication overhead (OH_t, % of communication resources used for model-parameter exchange).

The performance trends strongly suggest that predictive state augmentation is the main performance improvement source in the proposed approach. Despite using the same reward coefficients, migration techniques, and DRL hyperparameters in all experiments, PODRL benefits decrease with prediction quality.

5.3.6 Impact of Prediction Accuracy

We examine how prediction accuracy affects the performance of the proposed PODRL framework in order to better understand the role of predictive state augmentation. Prediction quality has a direct impact on offloading feasibility estimation, migration choices, and long-term policy optimization because the proposed approach depends on LSTM-based trajectory and channel forecasting.

Both trajectory and channel predictions were subjected to controlled Gaussian noise in order to assess robustness under imperfect prediction settings. From 0% to 20%, the prediction error level was progressively raised; higher error levels indicate more significant differences between the expected and real mobility/channel conditions.

The effect of forecast error on average end-to-end latency and deadline satisfaction rate is shown in Figure 15. Due to less dependable mobility-aware offloading decisions and more frequent migration adjustments, latency progressively rises as prediction accuracy declines. In a similar vein, poor forecasts might result in less-than-ideal executor selection and transient communication instability, which lowers deadline satisfaction.

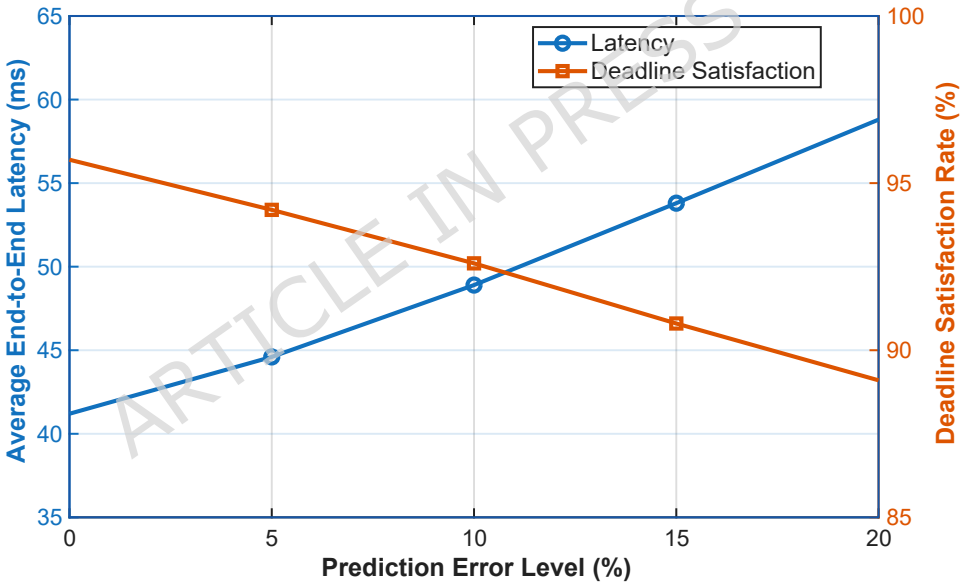


Figure 15. Impact of prediction error on average end-to-end latency and deadline satisfaction rate in the proposed PODRL framework.

5.3.7 Robustness Under Forecast Errors

Further stress-test experiments employing purposefully deteriorated trajectory and channel forecasts were carried out to assess the durability of the proposed PODRL architecture under poor prediction settings. This subsection examines more difficult real-world forecasting failures, such as random perturbations, stale predictions, and delayed forecast updates, in contrast to the experiments in Section 5.3.6, which examined gradual prediction accuracy degradation.

Four exemplary situations were examined:

- Precise prediction (normal prediction condition),
- Moderate forecast error (+5 dB SINR error and +5 m trajectory variation),
- Extreme forecast error (+15 dB SINR inaccuracy and +20 m trajectory variation),
- Predictions that are stale and have delayed update intervals.

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The performance of PODRL under these predicting settings is summarized in Table 8.

Table 8. Robustness of PODRL Under Forecast Errors

Scenario	Avg. End-to-End Latency (ms)	Energy Consumption (mWh/task)	Deadline Satisfaction (%)	Migration/Failure Rate (%)
Accurate Forecast	41.2 ± 2.1	51.6 ± 3.2	95.7	4.3
Mild Forecast Error (+5 dB / +5 m)	45.8 ± 2.4	55.3 ± 3.5	93.8	5.6
Severe Forecast Error (+15 dB / +20 m)	53.7 ± 2.9	63.1 ± 4.1	89.4	8.7
Stale Prediction (Delayed Updates)	57.4 ± 3.3	66.8 ± 4.5	87.9	10.2

5.3.8 Computational Runtime Analysis

We examine the average offloading decision runtime of PODRL and compare it with the primary DRL-based baselines in order to further assess the practical viability of the proposed PODRL architecture for real-time vehicular situations. The proposed framework is appropriate for activities operating at the top range of the evaluated latency requirements (e.g., 20–30 ms deadlines) and moderately delay-sensitive vehicular applications, according to Table 9's average decision time of 24.9 ms for PODRL. For ultra-low-latency safety-critical applications with stringent 10 ms deadlines, where the decision process might form part of the crucial latency path, this runtime should not be considered adequate. In order to further reduce inference latency, such cases might call for additional optimization strategies like model compression, decision precomputation, caching mechanisms, or more potent edge hardware.

Table 9. Computational Runtime Analysis of Offloading Decision Methods

Method	Avg. Decision Time (ms)	Real-Time Feasibility (10–30 ms Deadline)
VMECS [31]	8.7 ± 0.6	Yes
Multi-Agent DRL [22]	12.4 ± 0.9	Yes
Federated MA-DRL [27]	15.8 ± 1.1	Yes
Hybrid Hierarchical DDRL [29]	18.6 ± 1.3	Yes
Proposed PODRL	24.9 ± 1.7	Yes

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5.4. Discussion

The results of this study have to be interpreted in the context of a simulation-based system integration framework. Instead of proposing a novel theory of reinforcement learning, PODRL combines existing components, such as LSTM-based mobility and channel prediction, DDQN with dueling architecture and prioritized experience replay, predictive routing, proactive migration management, and federated learning. The main contribution is the explicit connection of anticipated future states with offloading decisions, action feasibility evaluation, reward shaping and migration control in a unified decision making framework.

The observed performance enhancements are due to several complementary causes. First, predictive mobility and channel awareness allow the framework to prevent offloading decisions that are expected to become infeasible owing to future connectivity degradation. Second, predictive transmission and migration strategies can eliminate needless retransmissions and handover-related disruptions, thus increasing both latency and energy efficiency under high-speed vehicular mobility. Third, priority-aware scheduling can achieve high deadline satisfaction rates even under large traffic loads by providing resources according to the job urgency and service requirements.

Results further show that incorporating future mobility and communication information into the selection process might bring significant advantages over solely reactive offloading solutions. In particular, the augmented MDP formulation enables the learning agent to concurrently consider the current resource conditions and the expected future network dynamics in choosing the execution locations. The ablation

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analysis reveals that the benefits presented are not due to a single module, but the result of a coordinated interplay between prediction, decision-making and migration mechanisms.

At the same time, several limits should be recognized. We tested the proposed framework solely through simulations, employing realistic vehicular movement traces and advanced 6G-inspired communication models. The simulation environment is enriched with advanced features like as terahertz communications, massive MIMO, and ultra-reliable low latency communication assumptions but it is not a deployed operational 6G-IoV infrastructure. Therefore, the published results serve as an indication of the potential effectiveness of prediction-coupled offloading schemes in controlled simulation conditions and not as a validation of an already mature real-world deployment.

Future works will include testing the framework with larger scale heterogeneous vehicle environments under more realistic communication uncertainties, and exploring deployment-oriented implementations on forthcoming IMT-2030 and beyond-5G vehicular edge computing platforms.

6. Conclusions

We proposed a prediction-coupled task offloading framework, PODRL, for forward-looking 6G-inspired vehicle edge computing environments in this paper. The proposed framework combines the conventional mobility prediction, wireless channel prediction, DRL, proactive migration management, and federated learning methods into a single decision-making architecture. PODRL does not introduce a fundamentally novel reinforcement learning algorithm, but rather highlights the system-level integration of prediction and decision-making modules for proactive work offloading in highly dynamic vehicular mobility scenarios.

The proposed methodology explicitly takes into account the projected future mobility and channel conditions in state representation, action feasibility determination, reward design and migration control. The system is able to provide more stable offloading decisions in fast varying vehicular situations by means of this prediction-aware coupling. Simulation results show consistent benefits in terms of latency, energy efficiency, deadline satisfaction and communication reliability compared to numerous reactive and learning-based baseline techniques, under the identical experimental settings.

The results also show that the use of anticipatory information for offloading and migration decisions can be practically beneficial in highly dynamic vehicle edge computing systems. In particular, we showed that the coordinated interaction among prediction, decision making, transmission management and federated learning components improves the overall system performance beyond the performance achievable by simply reactive offloading schemes.

6.1 Limitations and Future Work

However, despite the good results, many restrictions should be highlighted. First, the evaluation was performed in a fully simulated environment, based on actual vehicular mobility traces and forward-looking 6G-inspired communication models. Although these simulations are based on advanced assumptions for terahertz communications, massive MIMO and ultra-reliable low-latency communication, they cannot fully cover all the intricacies of future operational vehicle networks. Therefore, the performance benefits stated should be seen as evidence from the simulation, rather than validation of a deployed 6G-IoV system.

Second, the federated learning component shares model parameters rather than raw mobility or task data. However, potential privacy vulnerabilities in parameter-sharing mechanisms (e.g., model inversion or inference assaults) were not explicitly studied in this work. Third, the current implementation is based on simplified assumptions on the integrated sensing and communication capabilities and may not fully reflect the future IMT-2030 deployments.

In the future we plan to experimentally validate our proposals through large-scale vehicular edge computing testbeds, adopt more realistic communication and sensing models, enhance the privacy-preserving properties of federated learning via secure aggregation and differential privacy, and extend our solutions to cooperative multi-agent offloading scenarios with heterogeneous vehicles and infrastructure resources.

Acknowledgements:

The 2025 Hunan Provincial Humanities and Social Sciences Fund Project “Research on Innovative Design of Intangible Cultural Heritage Products in Southwest Hunan and Their Internationalization Pathways” (25YBA334). The 2025 Hunan Provincial Humanities and Social Sciences Fund Project “Research on the TV adaptation of Mao Dun Literature Prize-winning works in the new era” (25YBA378).

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Declarations

Ethical Approval Not applicable.

Competing interests The authors have no competing interests, or other interests that might be perceived to influence the results and/or discussion reported in this paper.

Authors' contributions The study was conceptualized and designed by all authors. Jiangjun Ruan and Mei Wang contributed to the conceptualization and methodology of the proposed framework. Lu Peng and Wentao Liu developed the algorithms and implemented the simulation environment. Peiling Zhou and Lianglin Xu performed data analysis and experimental validation. Man Ding supervised the research, contributed to technical refinement, and coordinated the overall study. Shouliang Lai provided guidance on the research design, reviewed the manuscript, and approved the final version for submission.

Funding: The authors did not receive any financial support for this study.

Availability of data and materials The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request. In addition, the full code, simulation scripts, and configuration files for reproducing the results presented in this manuscript are openly available at the Zenodo repository: https://doi.org/10.5281/zenodo.20324408. This includes the DDQN-based PODRL agent, LSTM predictors, federated aggregation modules, NS-3/TeraSim configuration templates, MATLAB figures, and sample mobility traces for experimental validation.

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